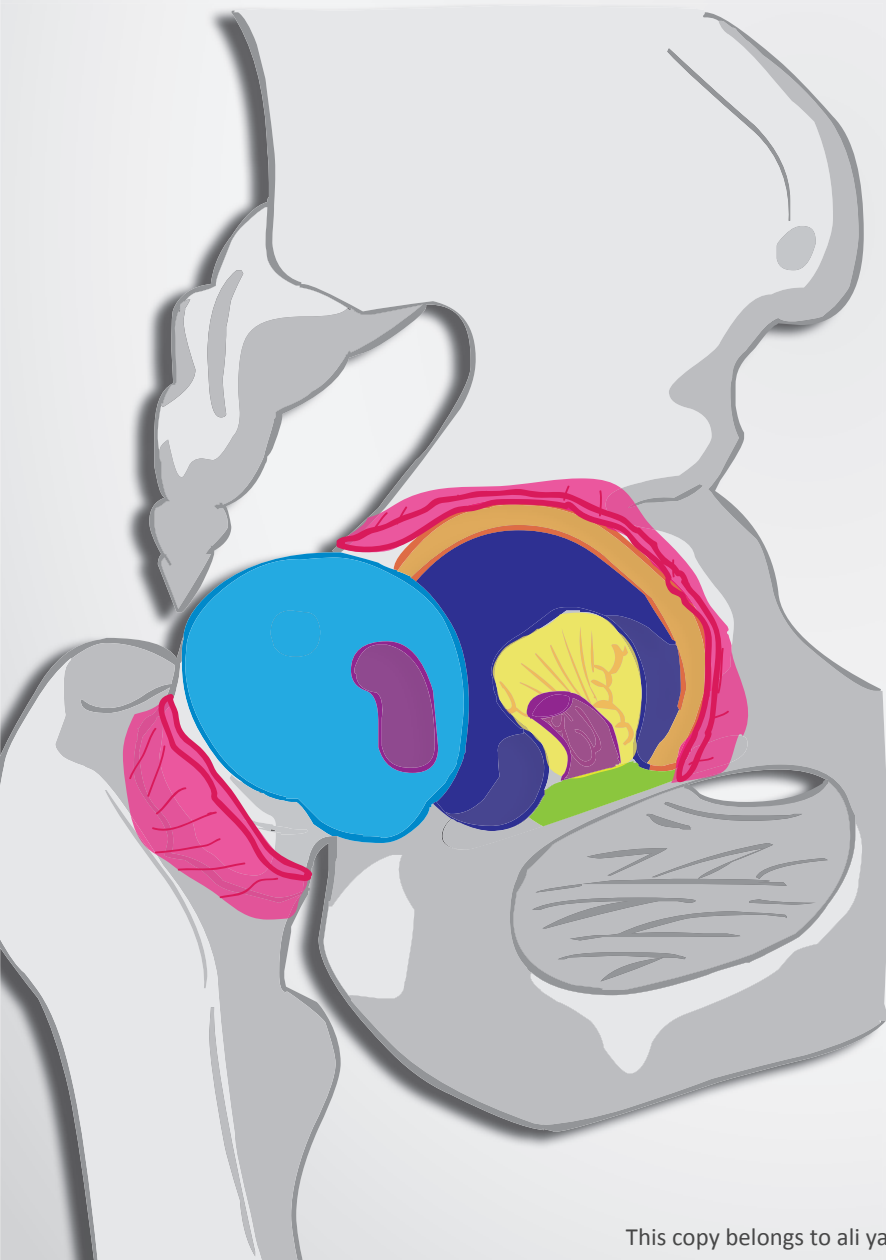


Memorix Anatomy

3 Joints

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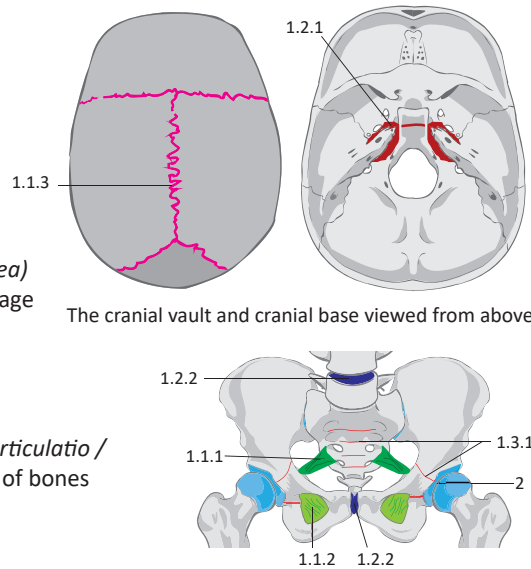


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Joints are connections between the bones of the skeleton. The two main types of joints are **synarthroses** and **diarthroses (synovial joints)**. In synarthroses, the opposing bone surfaces are connected to each other by connective tissue. The articulating surfaces in diarthroses are separated from each other by a space called the joint cavity.

Joints (*junctures*) – classification

- 1 **Synarthrosis** – a continuous connection of bones made by connective tissue
 - 1.1 **Fibrous joint** – a connection of bones through fibrous connective tissue
 - 1.1.1 **Syndesmosis**
 - 1.1.2 **Membrane (*membrana*)**
 - 1.1.3 **Suture (*sutura*)**
 - 1.2 **Cartilaginous joint (*junctura cartilaginea*)** – a connection of bones made by cartilage
 - 1.2.1 **Synchondrosis**
 - 1.2.2 **Symphysis**
 - 1.3 **Bony union (*junctura ossea*)**
 - 1.3.1 **Synostosis**
- 2 **Diarthrosis / synovial joint (*diarthrosis / articulatio / junctura synovialis*)** – a mobile connection of bones



The cranial vault and cranial base viewed from above

Fibrous joint (*junctura fibrosa*)

- 1 **Syndesmosis** – a connection of bones through fibrous connective tissue
 - 1.1 **Tibiofibular syndesmosis (*syndesmosis tibiofibularis*)**
 - 1.2 **Ligament (*ligamentum*)** – a band of fibrous connective tissue connecting one bone to another
 - 1.2.1 **Collagenous fibrous tissue** – makes up most ligaments
 - 1.2.2 **Elastic fibrous tissue** – makes up the ligamenta flava of the vertebral column
 - 1.3 **Gomphosis / socket** – the joint between the dental alveolus and tooth root
- 2 **Membranes (*membranae*)** – flat fibrous sheaths attached to bones
 - 2.1 **Interosseous membrane of forearm and leg and obturator membrane (*membrana interossea antebrachii et cruris et membrana obturatoria*)**
- 3 **Sutures (*suturæ*)** – fibrous joints between the bones of the skull
 - 3.1 **Serrate suture (*suturæ serrata*)** – comprises most sutures of the skull
 - the articulating surfaces are jagged like a saw and interdigitate with each other
 - 3.1.1 **Limbois suture (*sutura limbosa*)** – a type of serrate suture
 - the interdigitating processes of the suture overlap each other
 - found between the frontal and parietal bones
 - 3.1.2 **Denticulate suture (*sutura denticulata*)** – a type of serrate suture
 - the interdigitating processes of the suture are tooth shaped
 - located between the two parietal bones
 - 3.2 **Squamous suture (*sutura squamosa*)** – found between the temporal and parietal bones
 - 3.3 **Plane suture (*sutura plana*)** – found between the horizontal plates of the palatine bones
 - 3.4 **Schindylesis** – found between the vomer and sphenoidal rostrum

Cartilaginous joint (*junctura cartilaginea*)

- 1 **Synchondrosis** – a connection of bones by hyaline cartilage
 - 1.1 **Costochondral synchondroses (*synchondroses costochondrales*)**
 - are located between the ribs and costal cartilages
 - 1.2 **Cranial synchondroses (*synchondroses cranii*)** – connections of bones of the cranial base
 - 1.3 **Epiphyseal plates (*laminae epiphysiales*)** – the growth plates of long bones
- 2 **Symphysis** – a connection of bones by fibrous cartilage in regions of high pressure
 - 2.1 **Pubic symphysis (*symphysis pubica*)** – is located between the bodies of the pubic bones
 - 2.2 **Intervertebral discs (*disci intervertebrales*)** – are located between the vertebrae
 - 2.3 **Xiphosternal and manubriosternal joint (*symphysis xiphosternalis et manubriosternalis*)**

Bony union (*junctura ossea*)

- 1 **Synostoses** – connections of bone by bone, develop secondarily to another type of synarthrosis
 - 1.1 **(Intra)membranous synostoses** – develop from fibrous joints, occurs in the cranial sutures
 - 1.2 **Enchondral synostoses** – develop from cartilaginous joint
 - occurs in the acetabulum and in all epiphyseal plates

Arthron is the Greek term for joint.
Arthrology is the study of joints.

Abbreviations of some joints:

SC – sternoclavicular joint
AC – acromioclavicular joint
GH – glenohumeral joint
PIP – proximal interphalangeal joint
DIP – distal interphalangeal joint
SI – sacroiliac joint

The anatomical position is an upright standing position with the upper limbs hanging alongside the body with the palms facing forward. Joints are described in the anatomical position.

It is best to fix joints in the middle position when treating fractures, unless the type of injury needs a special type of fixation.

The blood supply of joints is provided by an arterial plexus (*rete articulare*).

Innervation of joints is usually provided by the same nerves which innervate the muscles that move the joint.

Arthrodia is an obsolete term for a freely moving ball and socket joint.

Enarthrosis is an obsolete term for cotyloid joint.

Amphiarthroses are slightly movable joints. The articulating surfaces are flat with a degree of incongruity that prevents all but slight sliding movements. The sacroiliac joint and the calcaneo-cuboid joint are examples of amphiarthroses.

Facet joint is a clinical term for a plane joint.

Clinical notes

Osteoarthritis is a degenerative joint disease that has a predilection for the large weight bearing joints. The most commonly affected joints are the hip (*coxarthrosis*), knee (*gonarthrosis*), shoulder (*omarthrosis*) and vertebral column (*spondylarthrosis*).

Rheumatoid arthritis is an inflammatory systemic autoimmune disease that can affect any of the joints of the human body. The synovium is primarily affected, but as the disease progresses the cartilage and underlying bone are also damaged. This disease has to be treated by permanent suppression of the immune system.

Joint-play is a passive movement in joints that cannot be performed actively. These are reciprocal shifts of articular surfaces, slight rotations and distractions.

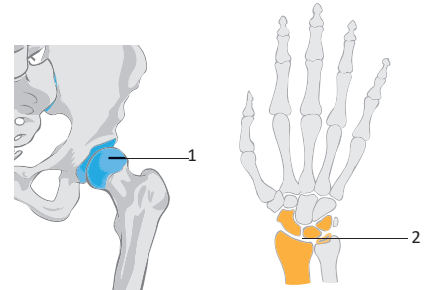
A synovial joint is a **connection between two or more bones** that **allows a significant degree of mobility**. The **articulating surfaces** in synovial joints are **covered by hyaline cartilage** and the joint is covered by an **articular capsule**. Under the capsule and between the two articulating surfaces lies a space called the **joint cavity**, which is filled with synovial fluid.

Synovial joint – description

- 1 **Type** – synovial joints are classified according to the number of articulating bones and the presence of articular discs or menisci into simple, compound and complex joints
- 2 **Shape** – is determined by the shape of the articular surfaces, which determines the range and type of movements
- 3 **Articular surface** (*facies articulares*) – can be flat, convex or concave
 - 3.1 **Articular head** (*caput articulare*) – is convex
 - 3.2 **Articular fossa** (*fossa articularis*) – is concave
- 4 **Articular capsule** (*capsula articularis*) – is composed of an internal synovial layer that supplies articular surfaces with nutrients and an external fibrous layer that provides mechanical stability to the joint
- 5 **Ligaments** – are found within the joint, outside the joint or within the articular capsule
 - limit movements and stabilise joints
- 6 **Accessory features** – specialised structures that function to improve the mobility and stability of the joint
 - specialised structures that compensate incongruity of articular surfaces, enlarge articular fossa, reduce attrition or prevent impaction of articular capsule
- 7 **Movements / range of motion** – describes in degrees the extent of a movement in a particular axis of a joint
 - 0° indicates a joint in the neutral position
- 8 **Neutral/middle position** – the position of the joint that has the least tension in the articular capsule

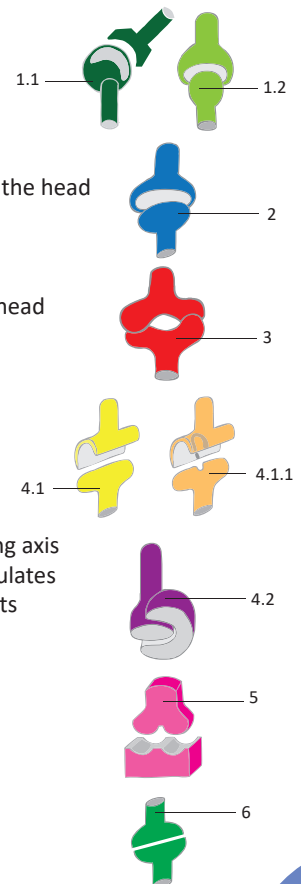
Types of synovial joints

- 1 **Simple joints** (*articulatio simplex*) – an articulation of two bones in the absence of articular discs and menisci
- 2 **Compound and complex joints**
 - 2.1 **Compound joints** – articulations of more than two bones
 - include the elbow joint
 - 2.2 **Complex joints** – indicates the presence of an articular disc or meniscus
 - include the wrist joint
 - some joints can be both complex and compound



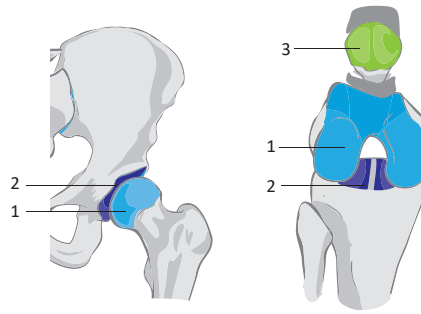
Shapes of synovial joints

- 1 **Ball-and-socket joint / spheroidal joint** (*articulatio spherioidea*)
 - allows rotational movements in three orthogonal axes
 - the articular head has a shape of a ball and the articular fossa resembles a socket
 - 1.1 **Free** – the surface of the articular fossa is smaller than the surface of the articular head
 - allows a large range of joint motion and is represented by the shoulder joint
 - 1.2 **Limited/cotyloid joint** (*articulatio cotylica*) – the surface of the fossa is deeper than the surface of the head
 - allows a limited range of joint motion and is represented by the hip joint
- 2 **Ellipsoid/condylar joint** (*articulatio ellipsoidea*)
 - the articular surfaces are ellipse-shaped (biconvex head and biconcave fossa)
 - allows rotational movements in the long axis of the joint and lateral sliding movements of the articular head
 - includes the wrist joint and the atlanto-occipital joint
- 3 **Saddle joint** (*articulatio sellaris*) – the articular surfaces resemble a horse rider sitting on a saddle
 - allows movements in two planes, includes the carpometacarpal joint of the thumb
- 4 **Cylindrical joint** (*articulatio cylindrica*) – has the shape of a cylinder
 - allows movements in one plane
 - 4.1 **Hinge joint** (*ginglymus*) – the axis of movement is perpendicular to the long axis of the bone
 - 4.1.1 **Trochlear joint** (*articulatio trochlearis*) – is cylinder-shaped
 - one articular surface possess a long cylindrical fossa with a ridge perpendicular to its long axis
 - the other articular surface has a small perpendicular indentation in which the ridge articulates
 - movements to the side are restricted by the shape of the joint and by collateral ligaments
 - includes the interphalangeal joints and the humero-ulnar joint
 - 4.2 **Pivot joint** (*articulatio trochoidea*) – a pivot-shaped articular surface that rotates in its long axis around a ring-shaped articular surface
 - includes the proximal and distal radio-ulnar joint and the median atlanto-axial joint
- 5 **Bicondylar joint** (*articulatio bicondylaris*) – has an articular head with two condyles
 - movements are possible in three axes and is represented by the temporomandibular joint
- 6 **Plane joint** (*articulatio plana*) – has flat articular surfaces, which allow sliding movements
 - usually has strong ligaments that limit the extent of joint motion
 - includes the acromioclavicular joint, femoropatellar joint and zygapophysial joint



Articular surfaces (*facies articulares*)

- 1 **Articular head** (*caput articulare*) – is convex
- 2 **Articular fossa** (*fossa articularis*) – is concave
- 3 **Articular surface** (*facies articularis*)
 - can be divided by grooves and ridges into a number of smaller articular facets

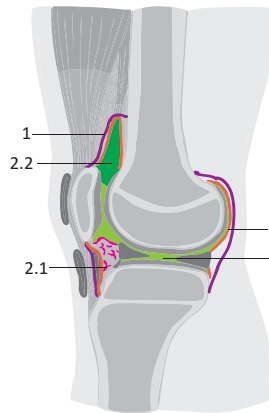


Articular cartilage

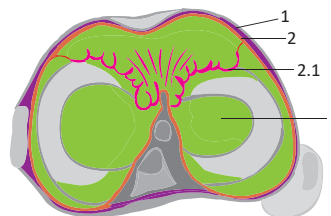
- covers the articular surfaces of joints
- composed of hyaline cartilage in most joints but can be composed of fibrocartilage in joints exposed to great stress, such as the sacro-iliac and sternoclavicular joints

Articular capsule (*capsula articularis*)

- the articular capsule surrounds the articular cavity
- it is attached to the circumference of the articular surfaces of the articulating bones
- 1 **Fibrous membrane/layer** (*membrana fibrosa / stratum fibrosum*)
 - the external layer of the joint capsule
 - provides mechanical protection to the joint
 - composed of dense connective tissue
 - often thickened by articular ligaments
- 2 **Synovial membrane/layer** (*membrana synovialis / stratum synoviale*)
 - the internal layer of the joint capsule
 - produces synovial fluid
 - supplies the joint cartilage with nutrients
 - composed of a layer of synoviocytes resting on an outer layer of loose connective tissue
 - 2.1 **Synovial fold** (*plica synovialis*)
 - a fold of the synovial membrane
 - the surface of the fold is divided into small projections called synovial villi (*villi synoviales*)
 - 2.2 **Articular recess** (*recessus articularis*)
 - a recesses within the articular cavity lined
- 3 **Articular/joint cavity** (*cavitas articularis*)
 - a space filled with synovial fluid, located between the articular surfaces and the synovial membrane



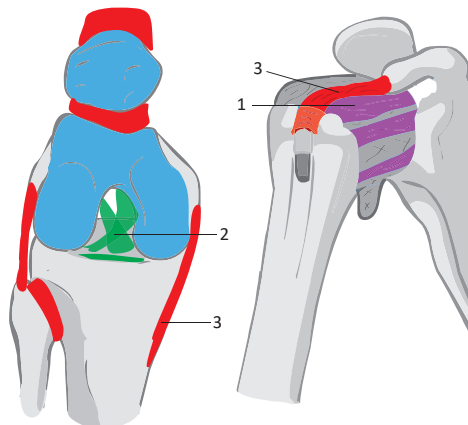
Sagittal section of the knee joint



Transverse section of the knee joint

Ligaments

- ligaments strengthen the articular capsule and stabilise connections between bones
- they limit the range of joint motion
- 1 **Capsular ligaments** (*ligamenta capsularia*)
 - part of the fibrous membrane of the articular capsule
 - include the glenohumeral ligaments of the shoulder joint
- 2 **Intracapsular ligaments** (*ligamenta intracapsularia*)
 - run inside the joint between the fibrous membrane and the synovial membrane
 - include the cruciate ligaments of the knee joint
- 3 **Extracapsular ligaments** (*ligamenta extracapsularia*)
 - run outside the joint
 - present in all joints in the human body



Open knee joint

Oblique view of the shoulder joint

Synovial bursae (*bursae synoviales*) are cavities lined with synovium and filled with synovial fluid. They act as cushions to prevent compression of muscles and ligaments in areas of higher stress. Some bursae may communicate with articular cavities although most do not.

Meniscoid folds / meniscoids are prominences of synovial membrane filled with connective tissue. One end is firmly connected to the joint capsule and the other moves freely in the articular cavity. Meniscoid folds cover surface bumps on articular surface to maintain a flat smooth articular surface. Meniscoid folds occur in the zygapophysial joints and the small joints of the hands and feet.

Articular recesses (*recessus articularis*) are synovial membrane-lined pouches extending outwards from the joint capsule. According to some theories, they can cause functional joint restriction.

Synovial fluid (*synovia*) is a translucent yellowish fluid produced by synovial cells (*synoviocytes*). Synovial fluid enables the articular surfaces to slide smoothly across each other with minimal resistance. It consists of plasma filtrate, hyaluronic acid, leukocytes and macrophages and nourishes the avascular parts of articular cartilage. In large joints, its volume is approximately 2–4 ml.

The tendon of the long head of the biceps brachii, which runs through the shoulder joint is, also an accessory of the joint.

Clinical notes

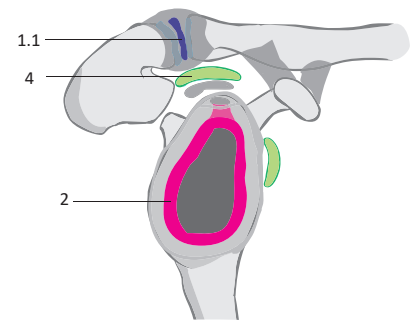
Joint restriction is a term given to reduced range of motion at a joint. It can have specific causes such as joint mice stuck between the articular surfaces or, in the case of functional joint restriction, have no fully elucidated cause.

Loose bodies / joint mice (*mus intraarticularis*) are small pieces of cartilage or bone that have broken off from the articular surface and are freely moving in the joint cavity. They typically occur in osteoarthritis and osteochondrosis dissecans. Loose bodies can prevent the joint from moving, causing joint restriction.

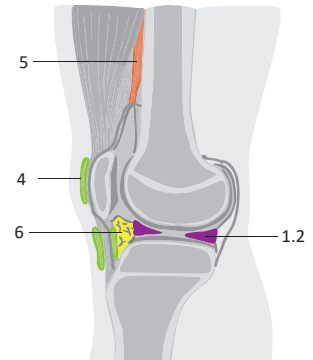
Bursitis is a term used for painful inflammation of synovial bursa that is not caused by infection nor an autoimmune process. It may originate after long-term overloading, high pressure or after injury. **Purulent bursitis** is a term used for infectious inflammation of synovial bursa.

Accessory features of synovial joints

- 1 **Discs and menisci** (*disci et menisci*) – are composed of fibrocartilage
 - are located in joint cavities between the articular surfaces
 - compensate the incongruity of the articular surfaces
 - 1.1 **Articular discs** (*discus articularis*) – divide the articular cavity into two parts
 - are found in the sternoclavicular, temporomandibular and radiocarpal joint
 - 1.2 **Menisci** (*meniscus articularis*)
 - are crescent-shaped structures that only partially divide the joint cavity
 - are found in the knee joint
- 2 **Labrum** (*labrum articulare*)
 - is composed of fibrocartilage and forms an elevated margin on the articular surface
 - enlarges the articular cavity in the shoulder joint and the hip joint
- 3 **Fibrocartilage** (*fibrocartilago*) – enlarges the surface of the articular fossa
 - is found in metacarpophalangeal, metatarsophalangeal, interphalangeal and talocalcaneonavicular joints
- 4 **Synovial bursae** (*bursae synoviales*) – cavities lined by the synovial membrane and containing synovial fluid
 - may be connected to a joint cavity, but the majority are not
- 5 **Articular muscles** (*musculi articulares*) – small muscles that attach to the articular capsule
 - prevent damage of articular capsule and provide dynamic stabilisation of joints
 - play a significant role in the elbow joint and knee joint
- 6 **Articular fat pads** (*corpora adiposa*) – fill the incongruities of articular surfaces
 - are found in the knee joint, hip joint and elbow joint
- 7 **Synovial / capsular fold** (*plica synovialis*) – synovial fold of the articular capsule inside the cavity
 - is found in the knee and elbow joints
- 8 **Recess** (*recessus*) – a pouch of the articular cavity, is found e.g. in the knee or elbow joint



Lateral view of the shoulder joint and coracoacromial ligament

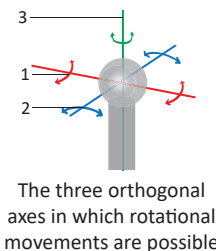


Sagittal section of the knee joint

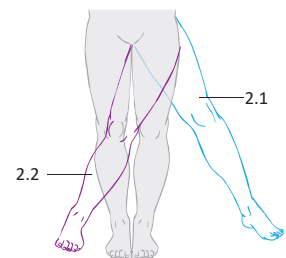
Movements

Main movements:

- 1 **Flexion and extension** – movements in the sagittal plane around the transverse axis
 - 1.1 **Flexion / ventral flexion** (*flexio*)
 - 1.2 **Extension** (*extensio*)
- 2 **Abduction and adduction**
 - movements in the frontal plane around the sagittale axis
 - 2.1 **Abduction** (*abductio*) – away from the midline
 - 2.2 **Adduction** (*adductio*) – towards the midline
 - 2.2.1 **Hyperadduction** – movements over the midline to the contralateral side
- 3 **Rotation** – movement around the long (lingitudinal/vertical) axis of the bone
 - 3.1 **External/lateral rotation** (*rotatio externa*)
 - 3.2 **Internal/medial rotation** (*rotatio interna*)

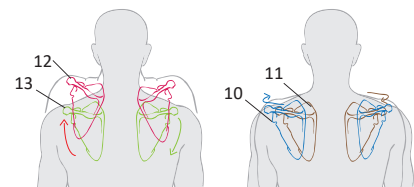
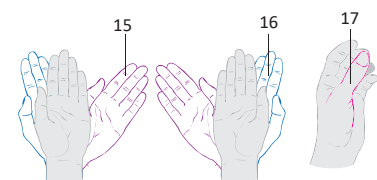
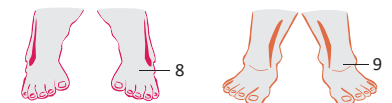
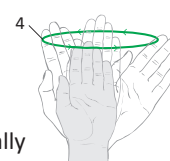
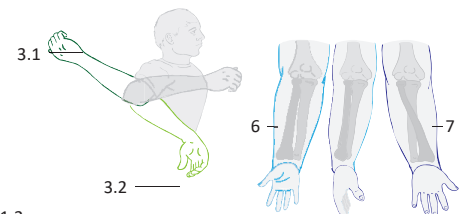
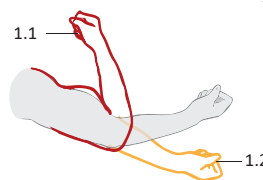


The three orthogonal axes in which rotational movements are possible



Other movements:

- 4 **Circumduction** (*circumductio*) – a combination of flexion, abduction, extension and adduction
- 5 **Torsion** (*torsio*) – rotational movements of two structures in opposite directions
 - occurs in the vertebral column
- 6 **Supination** (*supinatio*)
 - rotation of the forearm and hand until the hand palm points ventrally
 - the radius and ulna lie parallel to each other in complete supination
- 7 **Pronation** (*pronatio*)
 - rotation of the forearm and hand that results in the palm pointing dorsally
 - the radius crosses over the ulna during pronation
- 8 **Inversion** (*inversio*) – combination of flexion, adduction and supination of the foot
- 9 **Eversion** (*eversio*) – combination of extension, abduction and pronation of the foot
- 10 **Protraction** (*protractio*) – forward movement of the scapula or mandible
- 11 **Retraction** (*retractio*) – backward movement of the scapula or mandible
- 12 **Elevation** (*elevatio*) – upward movement of the scapula or mandible
- 13 **Depression** (*depressio*) – downward movement of the scapula or mandible
- 14 **Lateroflexion** (*lateroflexio*) – flexion of the vertebral column to the side
- 15 **Ulnar duction** (*ductio ulnaris*) – medial flexion of the hand
- 16 **Radial duction** (*ductio radialis*) – lateral flexion of the hand
- 17 **Opposition** (*oppositio*) – movement of the thumb towards the other fingers
- 18 **Reposition** (*repositio*) – movement of the thumb away from the other fingers



In the skull, **all the main types of joints are found**. The bones of the cranial vault are connected by **sutures**, which, during course of aging, change to **synostoses**. The majority of the bones of the base of the skull are connected to each other by **synchondroses**. The mandible is connected to the temporal bone by a **synovial joint**.

Synarthroses of the skull

- 1 **Sutures** – fibrous joints of the cranial vault
– there are 33 sutures in the skull

- 1.1 **Sagittal suture** (*sutura sagittalis*)
- 1.2 **Coronal suture** (*sutura coronalis*)
- 1.3 **Lambdoid suture** (*sutura lambdoidea*)
- 1.4 **Squamous suture** (*sutura squamosa*)

- 2 **Fontanelles** (*fonticuli cranii*) – joints of the skull of a newborn
– soft fibrous tissue that covers gaps between the cranial bones

3 Ligaments

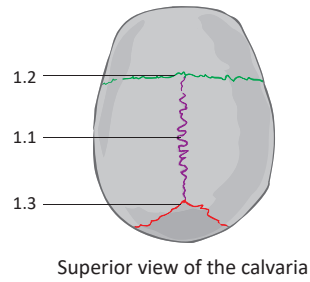
- 3.1 **Stylohyoid ligament** (*ligamentum stylohyoideum*)
- 3.2 **Sphenomandibular ligament** (*ligamentum sphenomandibulare*)
- 3.3 **Pterygospinous ligament** (*ligamentum pterygospinale*)
- 3.4 **Pterygomandibular raphe** (*raphe pterygomandibularis*)

- 4 **Gomphosis** – joints between dental alveoli and the roots of teeth

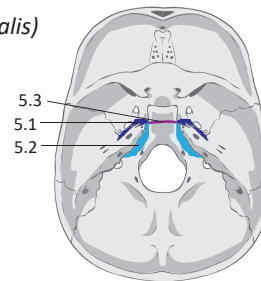
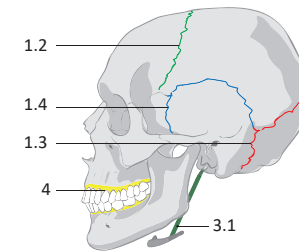
- 5 **Synchondroses** – firm continuous connections between bones that are mediated by hyaline cartilage

- 5.1 **Sphenopetrosal synchondrosis** (*synchondrosis sphenopetrosa*)
- 5.2 **Petro-occipital synchondrosis** (*petro-occipital synchondrosis*)
- 5.3 **Spheno-occipital synchondrosis** (*synchondrosis sphenooccipitalis*)
– closes at the 18th–20th years of life
- 5.4 **Anterior and posterior intra-occipital synchondrosis** (*synchondrosis intraoccipitalis anterior et posterior*)
– closes at the 4th–7th years of life
- 5.5 **Spheno-ethmoidal synchondrosis** (*synchondrosis sphenoethmoidalis*)
– closes after the 10th year of life

- 6 **Synostoses** – direct connection of bones
– occur after closure of the sutures and synchondroses



Superior view of the calvaria



Superior view of the cranial base

Temporomandibular joint (*articulatio temporomandibularis*)

- 1 **Type**: complex

- 2 **Shape**: bicondylar

- 3 **Articular head**: the head of the mandible

- 4 **Articular fossa**: the mandibular fossa of the temporal bone

- 5 **Capsule**: loose

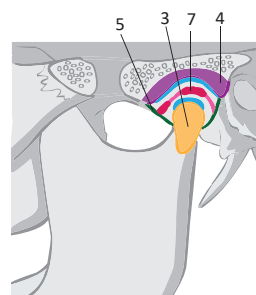
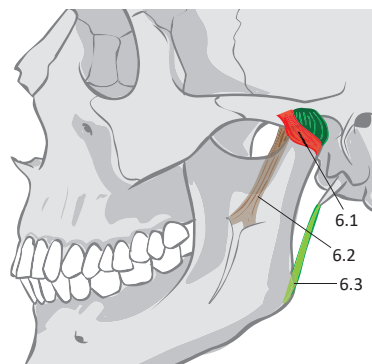
- 6 **Ligaments**:

- 6.1 **Lateral and medial ligaments** (*ligamentum laterale et mediale*)
- 6.2 **Sphenomandibular ligament** (*ligamentum sphenomandibulare*)
- 6.3 **Stylomandibular ligament** (*ligamentum stylomandibulare*)

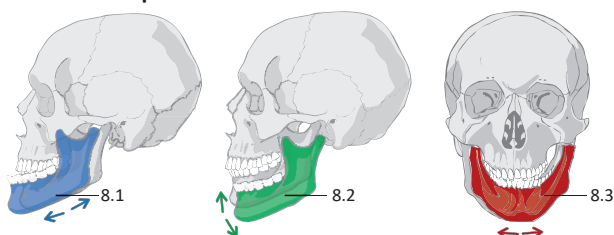
- 7 **Accessory features**: articular disc

- 8 **Movements**:

- 8.1 **Protraction and retraction**
- 8.2 **Depression and elevation**
- 8.3 **Lateropulsion**



Detail of the joint



The **fibrocartilage of the intervertebral discs** absorbs fluid during sleep. During wakefulness, it gradually releases this fluid. Therefore, humans are actually taller in the morning than in the evening. In adults, this difference in height can be as much as 2 cm.

The **nucleus pulposus** of the intervertebral disc is made of an incompressible fluid and behaves like a ball bearing.

In the majority of the population, the frontal suture fuses by the age of 2. However, 5–8 % of the population have a **persistent frontal suture** (*sutura frontalis persistens*, *sutura metopica*) separating the two frontal bones. Such individuals may have abnormally shaped foreheads.

Transverse occipital suture and sutura mendosa see page 22.
Fontanelles see page 35.

Lateropulsion and mediopulsion are terms for lateral excursions of the mandible in chewing. **Lateropulsion** is a movement of the mandible to the side. This movement occurs during chewing. The lateral pterygoid muscle working unilaterally produces these movements. **Mediopulsion** is a movement back to the middle position of the joint

Ligamenta interarcualia is an obsolete term for the yellow ligaments (*ligamenta flava*).

Clinical notes

Disorders of the temporomandibular joint cause pain in the temporal region. It is thus important to evaluate the temporomandibular joint in patients with unilateral headache originating in this region.

The **lateral pterygoid muscles** prevent the articular disc of the temporomandibular joint from becoming displaced and causing joint restriction. It is inserted through the capsule directly into the articular disc.

Spinal disc herniation (*prolapsed intervertebralis*) occurs when the anulus fibrosus tears and the nucleus pulposus protrudes through the tear. The herniating disc can impinge the spine nerve roots causing radiculopathy. For more details see the peripheral nervous system chapter.

In the vertebral column, all the main types of joints are found. **Fibrous joints** are represented by the long and short ligaments of the vertebral column. The intervertebral joints are a type of **cartilaginous joint**. The sacral and coccygeal vertebrae fuse together as **synostoses** during development to form the sacrum and coccyx, respectively. **Synovial joints** exist between the superior and inferior articular facets of adjacent vertebrae.

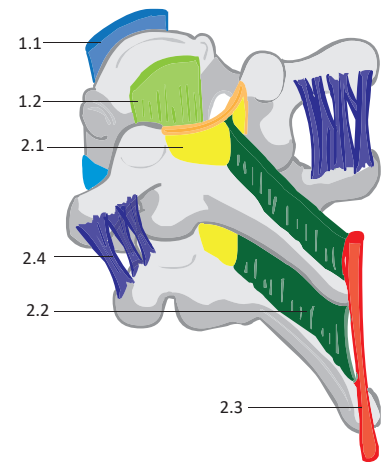
Syndesmoses

1 Long ligaments

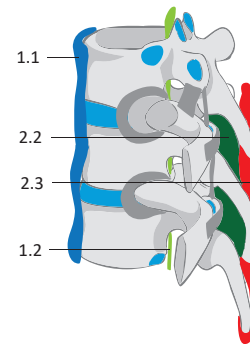
- **1.1 Anterior longitudinal ligament** (*ligamentum longitudinale anterius*)
 - is located on the ventral surface of the vertebral bodies C1 to S1
 - stabilises and fixates the vertebral column to the vertebral bodies
 - 1.1.1 **Anterior sacrococcygeal ligament** (*ligamentum sacrococcygeum anterius*)
 - the caudal continuation of the anterior longitudinal ligament
- **1.2 Posterior longitudinal ligament** (*ligamentum longitudinale posterius*)
 - is located on the dorsal surface of vertebral bodies inside the vertebral canal
 - prevents prolapse of intervertebral discs into the vertebral canal
 - 1.2.1 **Deep posterior sacrococcygeal ligament** (*ligamentum sacrococcygeum posterius profundum*)
 - the caudal continuation of the posterior longitudinal ligament
 - 1.2.2 **Tectorial membrane** (*membrana tectoria*)
 - the cranial continuation of the posterior longitudinal ligament
 - runs through the foramen magnum into the skull
 - covers the dorsal surface of the median atlanto-axial joint
- 1.3 **Superficial posterior sacrococcygeal ligament** (*ligamentum sacrococcygeum posterius superficiale*)
 - runs from the median sacral crest across the sacral and coccygeal horns to the posterior surface of the coccyx
 - borders the sacral hiatus

2 Short ligaments

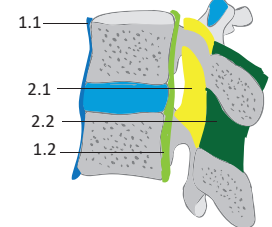
- **2.1 Ligamenta flava** (*ligamenta flava*) – are composed of elastic connective tissue
 - run between the vertebral arches
 - maintain an erect vertebral column
 - allow passive extension of the vertebral column
- **2.2 Interspinous ligaments** (*ligamenta interspinalia*)
 - run in between the spinous processes of adjacent vertebrae
- **2.3 Supraspinous ligament** (*ligamentum supraspinale*) – connects the tips of the spinous processes of the cervical, thoracic and upper lumbar vertebrae
 - 2.3.1 **Nuchal ligament** (*ligamentum nuchae*)
 - the cranial continuation of the supraspinous ligaments
 - runs from the spinous process of C7 to the external occipital protuberance
- **2.4 Intertransverse ligaments** (*ligamenta intertransversaria*)
 - are located between the transverse processes of the vertebrae



Two cervical vertebrae viewed obliquely from above, behind and left



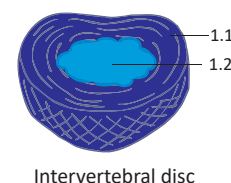
Three thoracic vertebrae viewed from the left



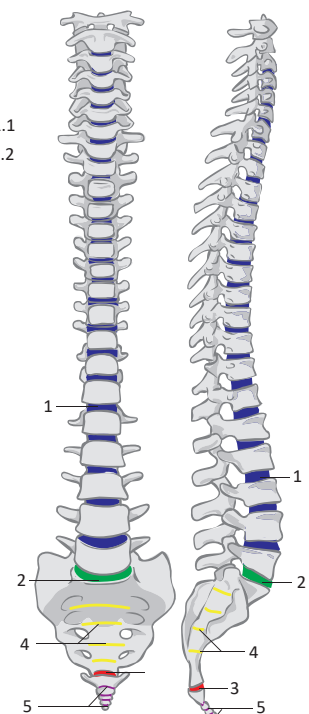
Sagittal section of two lumbar vertebrae

Synchondroses

- **1 Intervertebral discs** (*disci intervertebrales*) – 23 discs in total
 - evenly distribute pressure during movements of the vertebral column
 - are compressed by neighboring vertebrae to allow movements of the spine
 - **1.1 Anulus fibrosus** (*anulus fibrosus*) – a ring of fibrous cartilage
 - constitutes most of the mass of the intervertebral disc
 - **1.2 Nucleus pulposus** (*nucleus pulposus*)
 - a spherical incompressible gel-like substance located in the middle of the intervertebral disc
- **2 Lumbosacral symphysis** (*symphysis lumbosacralis*)
 - an intervertebral disc located between the vertebra L5 and the sacrum
 - is supported by the long and short ligaments of the vertebral column
- **3 Sacrococcygeal symphysis** (*symphysis sacrococcygea*)
 - an intervertebral disc located between the sacrum and coccyx that ossifies by the 40th year of life
 - during pregnancy, this joint becomes looser due to the effect of the hormone relaxin
 - is supported by the anterior, lateral and posterior sacrococcygeal ligaments, which are each divided into superficial and deep parts



Intervertebral disc



Synostoses

- **4 Synostoses of the sacrum**
- **5 Synostoses of the coccyx**

The **zygapophysial joints** are synovial joints connecting the superior and inferior articular processes of neighboring vertebrae. The **craniovertebral joints** consist of the joints connecting the occipital bone to the atlas and the joints between the atlas and axis.

Zygapophysial joints (*articulationes zygapophysiales*)

– there are 22 paired zygapophysial joints from the vertebrae C2 to S1

1 **Type:** simple

2 **Shape:** plane

3 **Articular surfaces:** the superior and inferior articular processes of neighboring vertebrae

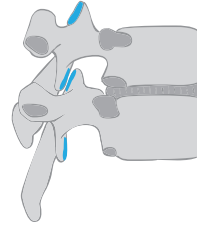
4 **Capsule:** is attached to the circumferences of the articular surfaces

5 **Ligaments:** long and short ligaments of the vertebral column

6 **Accessory features:**

6.1 **Meniscoid fold:** a fold of the synovial membrane filled with fibrous tissue

7 **Movements:** limited sliding movements



Atlanto-occipital joint (*articulatio atlantooccipitalis*)

– a paired joint between the atlas and occipital bone

1 **Type:** simple

2 **Shape:** ellipsoid

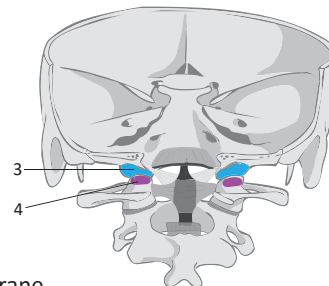
3 **Articular head:** the condyle of the occipital bone

4 **Articular fossa:** the superior articular process of the atlas

5 **Capsule:** is attached to the circumferences of the articular surfaces

6 **Ligaments:** the anterior and posterior atlanto-occipital membrane

7 **Movements:** anteflexion, retroflexion and a slight degree of lateral flexion



Posterior view of the base of the skull and the first two cervical vertebrae

Median atlanto-axial joint (*articulatio atlantoaxialis mediana*)

– an unpaired joint between the atlas and axis in the midline

1 **Type:** simple

2 **Shape:** pivot

3 **Articular head:** the dens of the axis

4 **Articular fossa:** the facet for the dens on the posterior surface of the anterior arch of the atlas

5 **Ligaments:**

5.1 **Cruciate ligament of atlas**

(*ligamentum cruciforme atlantis*)

● 5.1.1 **Transverse ligament of atlas**

(*ligamentum transversum atlantis*)

● 5.1.2 **Longitudinal bands**

(*fasciculi longitudinales*)

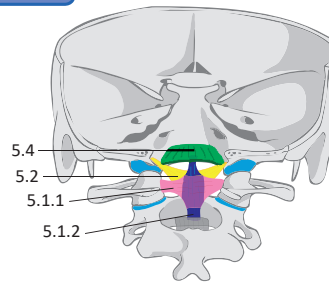
● 5.2 **Alar ligaments** (*ligamenta alaria*)

5.3 **Apical ligament of dens** (*ligamentum apicis dentis*)

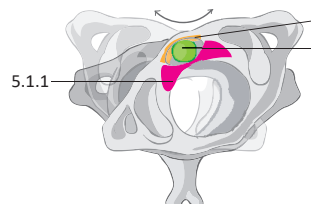
● 5.4 **Tectorial membrane** (*membrana tectoria*)

– covers the dorsal surface of the joint

6 **Movements:** 30° rotation of the atlas around the dens in each direction



Posterior view of the cranial base and the first two cervical vertebrae



Superior view of rotation of the atlas around the dens

Lateral atlanto-axial joint (*articulatio atlantoaxialis lateralis*)

– paired lateral articulations between the atlas and axis

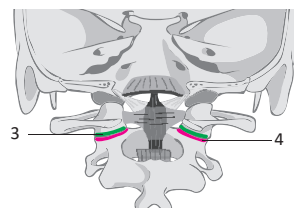
1 **Type:** simple

2 **Shape:** plane

3 **Head:** the inferior articular facet of the atlas

4 **Cavity:** the superior articular facet of the axis

5 **Movement:** very limited movements in all three axes – inclination of the atlas



Posterior view of the cranial base and the first two cervical vertebrae

The **atlanto-occipital joint** allows forward and backward movements of the head, which occur when nodding the head.

The **median atlanto-axial joint** allows rotatory movements, which occur when shaking the head.

Dorsiflexion is a clinical term for retroflexion of the vertebral column.

The **costotransverse foramen** (*foramen costotransversarium*) is a space between the neck of the rib and transverse process of the vertebra. It contains the costotransverse ligament.

The **radiate sternocostal ligaments** extend from the perichondrium of the ribs to the periosteum of the sternum and blend with the fibres from the contralateral radiate sternocostal ligaments, forming the **sternal membrane** (*membrana sterni*).

The **costoxiphoid ligaments** (*ligamenta costoxiphoidea*) extend from the sixth and seventh costal cartilages to the xiphoid process of the sternum.

We describe **rotational movements of the individual vertebrae**. However, when the whole spine rotates this is called torsion. Torsional movements require the vertebral segments to move in opposite directions to each other.

Clinical notes

Costochondritis is inflammation of the costochondral costal cartilages. It is a common cause of chest pain and can mimic acute coronary syndrome.

Rib restriction (restriction of the sternocostal joint) typically causes pain that is worsened by breathing movements.

Movements of the vertebral column

1 Anteflexion and retroflexion

- 1.1 **Cervical spine:** 0–50° ventrally and 0–70° dorsally
- 1.2 **Thoracic spine:** minimal, limited by the ribs
- 1.3 **Lumbar spine:** 0–25° ventrally and 0–90° dorsally

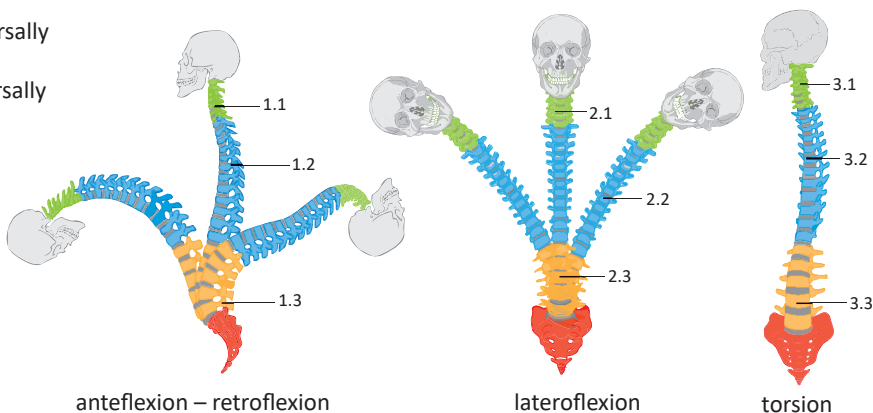
2 Lateroflexion

- 2.1 **Cervical spine:** 0–45°
- 2.2 **Thoracic spine:** minimal, limited by the ribs
- 2.3 **Lumbar spine:** 0–35°

3 Rotation (torsion)

- 3.1 **Cervical spine:** 0–70°
(0–30° in the median atlanto-axial joint)
- 3.2 **Thoracic spine:** 0–30°
- 3.3 **Lumbar spine:** 0–10° (provided in the discs)

4 Spring movements



anteflexion – retroflexion

lateroflexion

torsion

Thoracic joints – Juncturae thoracis

Movements of the ribs expand and contract the thorax, producing the inspiration and expiration movements of breathing. These movements are made possible by the **costovertebral joints**, which connect the ribs to the vertebral column, and by the **sternocostal joints**, which connect the ribs to the sternum. True ribs articulate with the sternum. The ventral ends of false ribs articulate with the rib above via interchondral joints. Floating ribs have no articulation at their ventral ends.

Synarthroses of the thorax

1 Syndesmoses

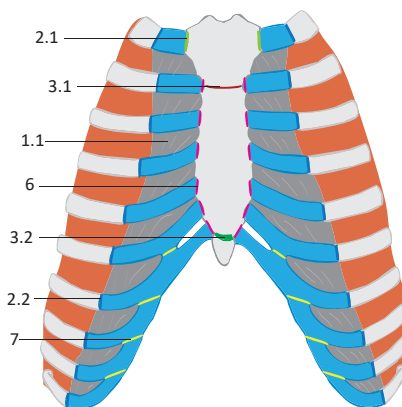
- 1.1 **External intercostal membrane** (*membrana intercostalis externa*)
 - the ventral continuation of the external intercostal muscles
 - is located in the ventral third of the intercostal space
- 1.2 **Internal intercostal membrane** (*membrana intercostalis interna*)
 - the dorsal continuation of the internal intercostal muscles
 - is located in the dorsal third of the intercostal space

2 Synchondroses

- 2.1 **Synchondrosis of first rib** (*synchondrosis costae primae*)
 - an articulation between the first rib and the manubrium of the sternum
- 2.2 **Costochondral synchondroses** (*synchondroses costochondrales*)
 - articulations between the ventral ends of the 1st–10th rib and their costal cartilages

3 Symphyses

- 3.1 **Manubriosternal joint** (*symphysis manubriosternalis*) – fuses by the 60th year of life
- 3.2 **Xiphosternal joint** (*symphysis xiphosternalis*) – fuses by the 40th–60th year of life



Costovertebral joints (*articulationes costovertebrales*)

● 4 Joint of head of rib (*articulatio capitis costae*)

- 4.1 **Articular head:** the head of the rib which is divided by a small crest into two articular facets
- 4.2 **Articular fossa:** the superior and inferior costal facets of the vertebral body
- 4.3 **Ligaments:** the radiate and intra-articular ligaments of the head of the rib
(*ligamentum capitis costae intraarticulare et ligamentum capitis costae radiatum*)

● 5 Costotransverse joint (*articulatio costotransversaria*)

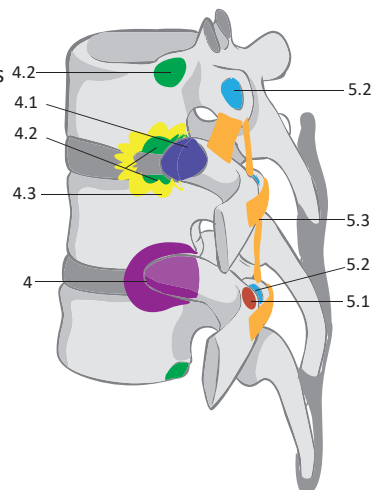
- 5.1 **Articular head:** tubercle of rib
- 5.2 **Articular fossa:** transverse costal facet
- 5.3 **Ligaments:** superior and lateral costotransverse ligaments
(*ligamentum costotransversarium superius et laterale*)

Sternocostal joints (*articulationes sternocostales*)

● 6 Sternocostal joint (*articulationes sternocostales*)

- 6.1 **Articular head:** cartilage of the 2nd–7th rib
- 6.2 **Articular fossa:** costal notches on the sternum for the 2nd–7th rib
- 6.3 **Ligaments:** the radiate sternocostal ligaments (*ligamenta sternocostalia radiata*) forming the sternal membrane (*membrana sterni*) and the intra-articular sternocostal ligaments (*ligamenta sternocostalia intraarticularia*)

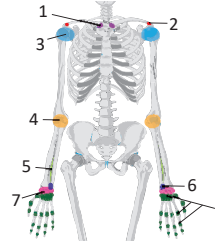
● 7 Interchondral joints (*articulationes interchondrales*) – between the cartilages of the 6th–9th rib



Three thoracic vertebrae viewed from the left

The vast majority of the joints of the upper limb are **synovial joints**. The radius and ulna, however, are connected to each other by a **fibrous interosseous membrane**.

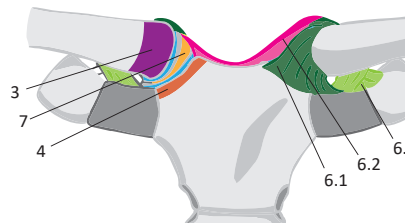
- 1 **Sternoclavicular joint** (*articulatio sternoclavicularis*)
- 2 **Acromioclavicular joint** (*articulatio acromioclavicularis*)
- 3 **Glenohumeral joint / shoulder joint** (*articulatio humeri/glenohumeralis*)
- 4 **Elbow joint** (*articulatio cubiti*)
- 5 **Radio-ulnar syndesmosis** (*syndesmosis radioulnaris*)
- 6 **Distal radio-ulnar joint** (*articulatio radioulnaris distalis*)
- 7 **Wrist joint** (*articulatio radiocarpalis*)
- 8 **Joints of hand** (*articulationes manus*)



5.1 Sternoclavicular joint – *Articulatio sternoclavicularis*

The sternoclavicular joint is a **complex joint** that allows only **very limited movements**. It cooperates with the other joints of the pectoral girdle.

- 1 **Type:** complex
- 2 **Shape:** ball-and-socket
- 3 **Articular head:** the sternal facet of the clavicle
- 4 **Articular fossa:** the clavicular notch on the sternum
- 5 **Capsule:** is attached to the circumferences of the articular surfaces
- 6 **Ligaments:**
 - 6.1 **Anterior and posterior sternoclavicular ligament** (*ligamentum sternoclaviculare anterius et posterius*)
– are connected to the capsule ventrally and dorsally, respectively
 - 6.2 **Interclavicular ligament** (*ligamentum interclaviculare*)
– dorsally interconnects the two clavicles
 - 6.3 **Costoclavicular ligament** (*ligamentum costoclaviculare*)
– connects the clavicle to the cartilage of the first rib
- 7 **Accessory features:** articular disc
- 8 **Movements:** very limited in all directions



Fornix humeri is a fibrous arch above the shoulder joint shaped by the coracoacromial ligament. Abduction of the arm greater than 90° is limited by the greater tubercle coming into contact with the fornix humeri. The arm can only be abducted further by concomitant external rotation of the scapula.

The shoulder is a **collective term for:**

- 1 The glenohumeral joint
 - 2 The acromioclavicular joint
 - 3 The sternoclavicular joint
 - 4 The functional connection between the scapula and thorax ("scapulothoracic joint")
- Movements of the shoulder are facilitated by movements at these four connections.

The **foramen of Weitbrecht** (*foramen ovale Weitbrechti*) is one of two weak spots in the glenohumeral joint capsule, located between the superior and middle glenohumeral ligaments.

The **blood supply of the shoulder joint** is provided mainly by the posterior humeral circumflex artery.

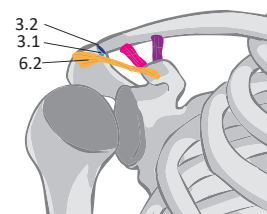
The **innervation of the shoulder joint** is provided by the suprascapular nerve.

Neutral position of the sternoclavicular and acromioclavicular joint corresponds to the anatomical position.

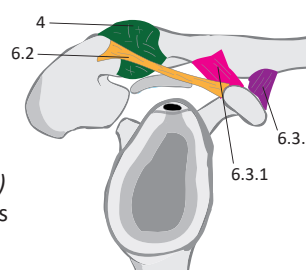
5.2 Acromioclavicular joint – *Articulatio acromioclavicularis*

The acromioclavicular joint is a **simple plane joint** that allows only a **limited degree of movement**. It variably contains an articular disc. It cooperates with the other joints of the pectoral girdle in producing movements of the upper limb.

- 1 **Type:** simple
- 2 **Shape:** plane
- 3 **Articular surfaces:**
 - 3.1 **Clavicular facet of the acromion** (*facies articularis clavicularis acromii*)
 - 3.2 **Acromial facet of the clavicle** (*facies articularis acromialis clavicularae*)
- 4 **Capsule:** firm, attached to the circumference of articular surfaces
- 5 **Accessories:** an articular disc is present in 50 % of the population
- 6 **Ligaments:**
 - 6.1 **Acromioclavicular ligament** (*ligamentum acromioclaviculare*)
– strengthens the cranial part of the joint
 - 6.2 **Coraco-acromial ligament** (*ligamentum coracoacromiale*)
– limits abduction of the arm above 90°
 - 6.3 **Coracoclavicular ligament** (*ligamentum coracoclaviculare*)
– limits movements of the scapula, composed of two parts
 - 6.3.1 **Trapezoid ligament** (*ligamentum trapezoideum*)
 - 6.3.2 **Conoid ligament** (*ligamentum conoideum*)
- 7 **Movements:** very limited in all directions



Anterior view of the right shoulder



Lateral view of the right shoulder

Clinical notes

Violent impacts transferred from the upper limb lead more often to a fracture of the clavicle than to a dislocation of the sternoclavicular joint, because of firmness of the articular capsule and ligaments.

Due to its small articular fossa and large articular head, the shoulder joint is **frequently dislocated**. An anterior inferior shoulder dislocation is the most frequent type of shoulder dislocation, occurring in more than 90 % of cases.

Shoulder impingement syndrome is inflammation of the tendons of the rotator cuff muscles typically caused by overuse of these muscles. The inflamed tendons swell up and movements of the shoulder become painful.

A **SLAP lesion** (*superior labral tear from anterior to posterior*) is an injury to the glenoid labrum, characterised by a tear of the uppermost part of the labrum adjacent to the insertion of the long head of the biceps brachii.

The shoulder joint is a ball-and-socket joint with the greatest **extent of movement** of all the joints of the human body. It allows **movements in all three planes**. The incongruity between the shallow **articular fossa** and the large articular head inflicts great instability on the joint. This is compensated by the **rotator cuff muscles** which attach to the head of the humerus and hold it firmly in the articular fossa of the scapula.

1 **Type:** simple

2 **Shape:** ball-and-socket

3 **Articular head:** the head of the humerus

4 **Articular fossa:** the glenoid cavity of the scapula

5 **Capsule:** attaches to the neck of the scapula and the anatomical neck of the humerus

5.1 **Axillary recess** (*recessus axillaris*)

– the caudal recess of the articular capsule

5.2 **Intertubercular synovial sheath** (*vagina synovialis intertubercularis*)

– the synovial sheath of the tendon of the long head of the biceps brachii

6 **Ligaments:**

6.1 **Coracohumeral ligament** (*ligamentum coracohumerale*) – runs from the coracoid process of the scapula to the greater tubercle of the humerus

6.2 **Glenohumeral ligaments** (*ligamenta glenohumeralia*) – three capsular ligaments – run through the fibrous layer of the articular capsule

6.2.1 **Superior glenohumeral ligament**

(*ligamentum glenohumerale superius*)

6.2.2 **Middle glenohumeral ligament**

(*ligamentum glenohumerale medium*)

6.2.3 **Inferior glenohumeral ligament**

(*ligamentum glenohumerale inferius*)

6.3 **Coraco-acromial ligament** (*ligamentum coracoacromiale*)

– lies above the joint between the coracoid process and acromion

6.4 **Transverse humeral ligament** (*ligamentum transversum humeri*)

– runs over the greater and lesser tubercle of the humerus

– bridges the intertubercular sulcus, holding the tendon of the long head of the biceps brachii firmly in place

7 **Accessory features:**

Inside the articular capsule:

7.1 **Glenoid labrum** (*labrum glenoidale*) – a cartilaginous margin that enlarges the articular cavity

7.2 **Tendon of long head of biceps brachii** (*tendo capitis longi musculi bicipitis brachii*)

– originates on the supraglenoid tubercle near the glenoid labrum and runs between the fibrous and synovial layers of the articular capsule

Outside the articular capsule:

7.3 **Subdeltoid bursa** (*bursa subdeltoidea*) – is located under the deltoid

7.4 **Subacromial bursa** (*bursa subacromialis*) – is located under the acromion

7.5 **Subtendinous bursa of subscapularis** (*bursa subtendinea musculi subscapularis*)

8 **Movements:** large range of joint motion in all three planes

8.1 **Ventral flexion:** 0–90°, with concomitant external rotation of the inferior angle of the scapula up to 180°

8.2 **Dorsiflexion (extension):** 0–50°

8.3 **Abduction:** 0–90°, with concomitant external rotation of the inferior angle of the scapula up to 180°

8.4 **Adduction (hyperadduction):** 0–75°

8.5 **External rotation:** 0–90°

8.6 **Internal rotation:** 0–90°

9 **Neutral position:** mild flexion and 45° of abduction

Rotator cuff

– the rotator cuff is a group of four muscles that attach to the greater and lesser tubercle of the humerus

– they stabilise the shoulder joint and aid in its rotation

1 **Supraspinatus** (*musculus supraspinatus*)

– inserts on the superior facet of the greater tubercle

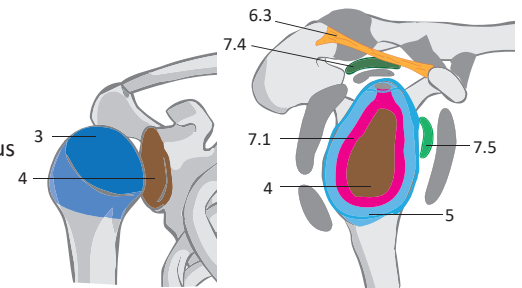
2 **Infraspinatus** (*musculus infraspinatus*)

– inserts on the middle facet of the greater tubercle

3 **Teres minor** (*musculus teres minor*)

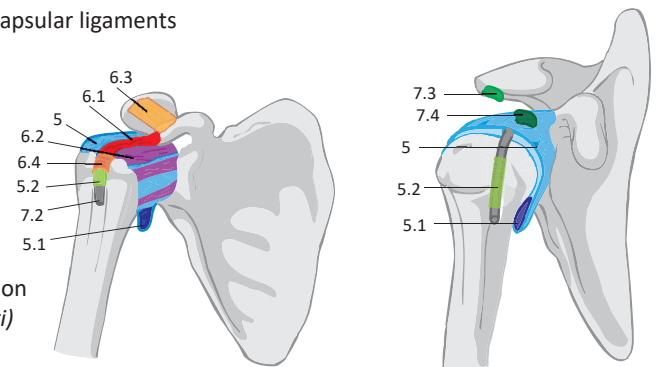
– inserts on the inferior facet of the greater tubercle

4 **Subscapularis** (*musculus subscapularis*) – inserts on the lesser tubercle



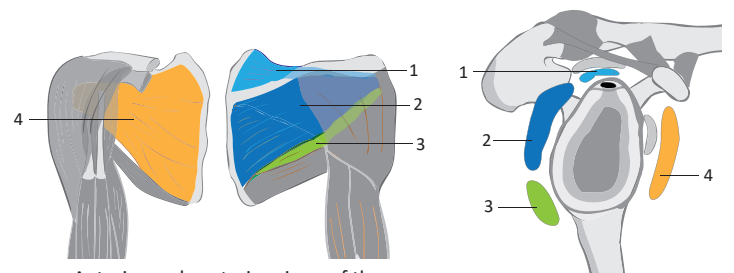
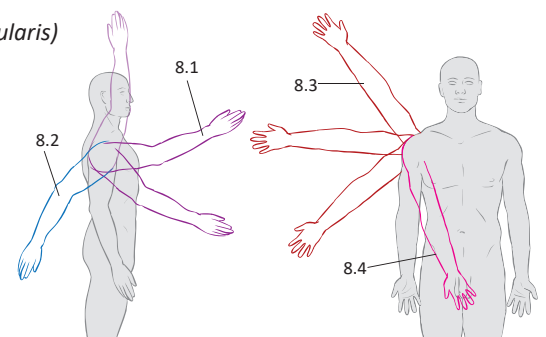
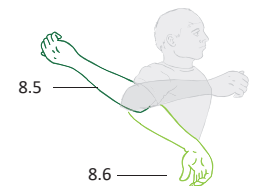
Anterior view of the right shoulder

Lateral view of the right shoulder



Anterior view of the right shoulder

Left oblique view of the right shoulder with the joint capsule partially opened



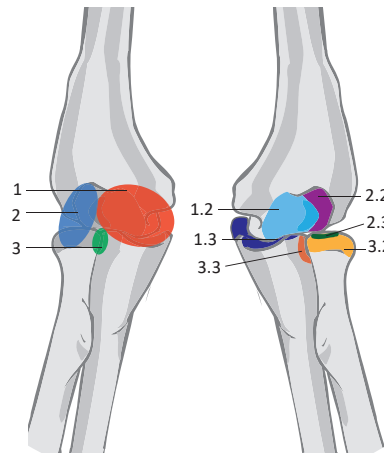
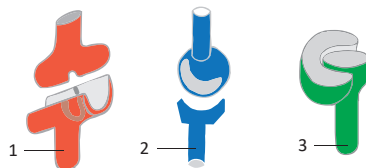
Anterior and posterior views of the muscles attaching to the scapula

Lateral view of the right shoulder

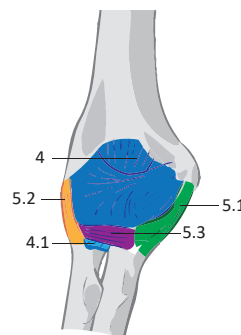
The elbow joint is a compound synovial joint made up of **three bones articulating in three separate simple joints**. It is a **biaxial joint** that allows flexion and extension and, in cooperation with the distal radio-ulnar joint, pronation and supination.

Joints comprising the elbow joint

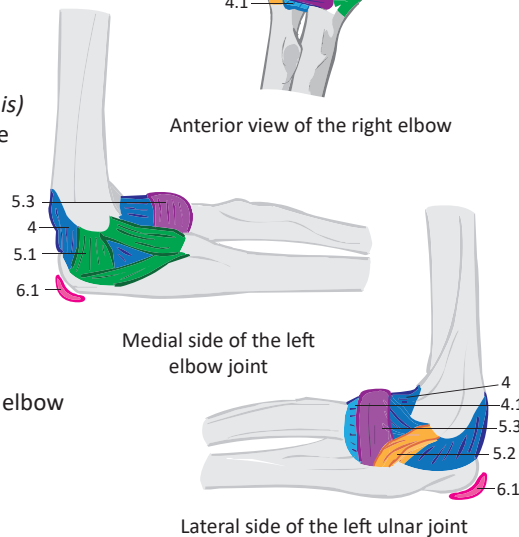
- **1 Humero-ulnar joint** (*articulatio humeroulnaris*)
 - 1.1 **Shape:** trochlear
 - 1.2 **Articular head:** the trochlea of the humerus
 - 1.3 **Articular fossa:** the trochlear notch of the ulna
- **2 Humeroradial joint** (*articulatio humeroradialis*)
 - 2.1 **Shape:** ball-and-socket
 - 2.2 **Articular head:** the capitulum of the humerus
 - 2.3 **Articular fossa:** the articular facet of the head of the radius
- **3 Proximal radio-ulnar joint** (*articulatio radioulnaris proximalis*)
 - 3.1 **Shape:** pivot
 - 3.2 **Articular head:** the articular circumference of the head of the radius
 - 3.3 **Articular fossa:** the radial notch of the ulna



Anterior view of the right and left elbow

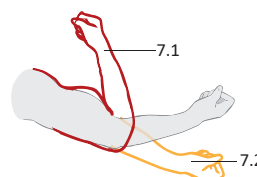
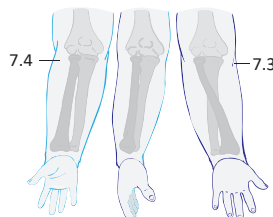


Anterior view of the right elbow



Medial side of the left elbow joint

Lateral side of the left ulnar joint



Common parts of the elbow joint

- **4 Capsule:** attaches to the humerus inferior to the epicondyles, to the neck of the radius and to the ulna on the borders of its articular surface
 - **4.1 Sacciform recess** (*recessus sacciformis*)
 - a pouch within the articular capsule located over the neck of the radius
- 5 Ligaments:**
 - **5.1 Ulnar collateral ligament** (*ligamentum collaterale ulnare*)
 - a triangular ligament on the medial side of the elbow
 - **5.2 Radial collateral ligament** (*ligamentum collaterale radiale*)
 - is located on the lateral side of the elbow
 - **5.3 Annular ligament of radius** (*ligamentum anulare radii*)
 - attaches to the ulna
 - encircles the head of the radius
- 6 Accessory features:**
 - **6.1 Subtendinous bursa of triceps brachii** (*bursa subtendinea musculi tricipitis brachii*)
 - in the place of the olecranon
- 7 Movements:**
 - **7.1 Flexion:** 0–145°
 - **7.2 Extension:** 0–5°
 - hyperextension up to 10° is physiological in females due to a smaller sized olecranon
 - **7.3 Pronation:** 0–90°
 - the radius is crossed over the ulna
 - **7.4 Supination:** 0–90°
 - the radius and ulna are parallel
- 8 Neutral position:** is in mild flexion and pronation

The **triangular fibrocartilage complex** (TFCC) is a clinical term used for the articular disc of the wrist joint and its associated ligaments, comprising the palmar and dorsal radioulnar ligaments and ulnocarpal ligaments (ulnolunate and ulnotriquetral ligaments). It separates the head of the ulna from the proximal row of the carpal bones. From the distal side, the disc looks like an isosceles triangle with the base adjacent to the ulnar notch of the radius and the apex at the styloid process of the ulna. The legs of the triangle grow together with the joint capsule of the distal radioulnar joint and form the palmar and dorsal radioulnar ligament, which help to stabilise the carpal bones.

The **ulnar collateral ligament** consists of **three ligaments, which form a triangle**:

- the humerocoronoid ligament (*lig. humerocoronoideum*)
- the olecranon humeral ligament (*lig. olecranon humerale*)
- the oblique ligament (*lig. obliquum*)

The **posterior interosseous vessels** (*vasa interossea posteriora*) pass through an opening between fibres of the oblique cord and the interosseous membrane to enter the dorsal compartment of the forearm.

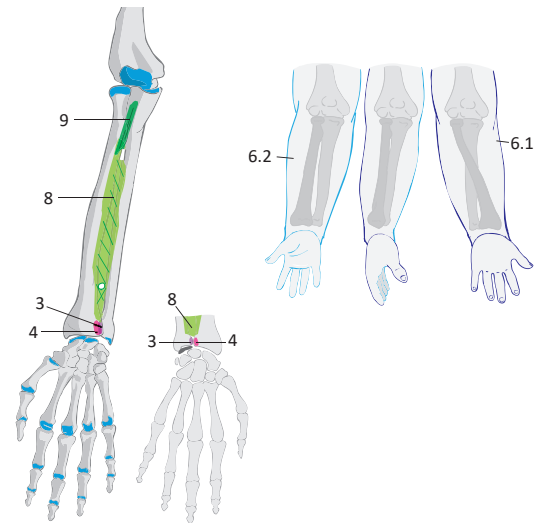
Clinical notes

The **anular ligament of the radius** is not fully developed in children until the age of 5 to 6. Under weight of the forearm, a subluxation of the humeroradial joint can occur. This condition is termed painful pronation (*pronatio dolorosa*).

The proximal and distal radio-ulnar joints form a **functional unit** that makes **pronation and supination** movements of the forearm possible. The bones of the forearm are united by a **syndesmosis**, which has two components: the proximal narrow **oblique cord** and the distal widened **interosseous membrane of the forearm**.

Distal radio-ulnar joint (*articulatio radioulnaris distalis*)

- 1 **Type:** simple
- 2 **Shape:** pivot
- 3 **Articular head:** the articular circumference of the head of the ulna
- 4 **Articular fossa:** the ulnar notch of the radius
- 5 **Capsule:** attached to the circumference of articular surfaces
- 6 **Movements:** are combined with movements of the proximal radio-ulnar joint
 - 6.1 **Pronation:** 0–90°
the radius crosses over ulna
the palm is turned to face dorsally
 - 6.2 **Supination:** 0–90°
the radius moves so that it lies parallel to the ulna
the palm is turned to face ventrally
- 7 **Neutral position:** is mild pronation



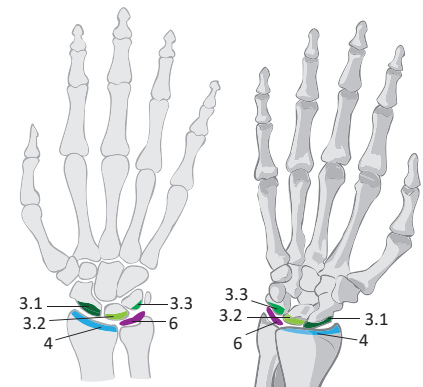
Anterior and posterior view of the right forearm and hand

Radio-ulnar syndesmosis (*syndesmosis radioulnaris*)

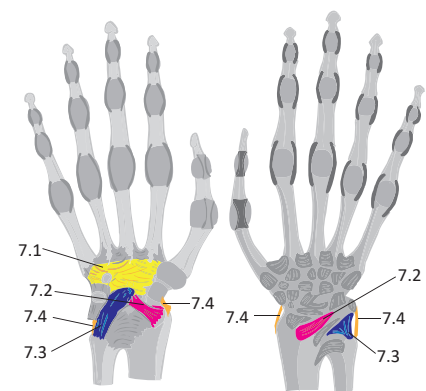
- 8 **Interosseous membrane of forearm** (*membrana interossea antebrachii*)
– is composed of fibers that run mediolaterally from the radius to the ulna
- 9 **Oblique cord** (*chorda obliqua*)
– a narrow band of fibers running laterodistally from the ulna to the radius

Composed of **four articulating bones** and containing an **articular disc**, the **ellipsoid shaped wrist joint** is classified as being both **complex and compound**. Allows movements in two planes, both separately and simultaneously.

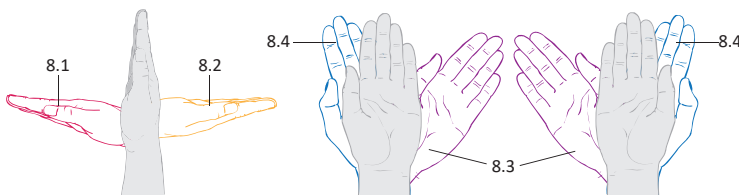
- 1 **Type:** both complex and compound
- 2 **Shape:** ellipsoid
- 3 **Articular head:**
 - 3.1 **Scaphoid**
 - 3.2 **Lunate**
 - 3.3 **Triquetrum**
- 4 **Articular fossa:** the carpal articular surface of the radius and the articular disc
- 5 **Capsule:** is loose and allows a wide range of movements
- 6 **Accessory features:** an articular disc, which forms the medial part of the articular fossa and articulates with the ulna (the ulna does not directly articulate with the carpal bones)
- 7 **Ligaments:**
 - 7.1 **Radiate carpal ligament** (*ligamentum carpi radiatum*)
 - 7.2 **Palmar and dorsal radiocarpal ligament** (*ligamentum radiocarpale palmare et dorsale*)
 - 7.3 **Palmar and dorsal ulnocarpal ligament** (*ligamentum ulnocarpale palmare et dorsale*)
 - 7.4 **Ulnar and radial collateral ligament of the wrist joint** (*ligamentum collaterale carpi ulnare et radiale*) – are weak ligaments
- 8 **Movements:**
 - 8.1 **Palmar flexion:** 0–80°
 - 8.2 **Dorsiflexion:** 0–85°
 - 8.3 **Ulnar deviation:** 0–45°
 - 8.4 **Radial deviation:** 0–15°
- 9 **Neutral position** – is in the direction of the long axis of the forearm



Anterior view of the left and right hand



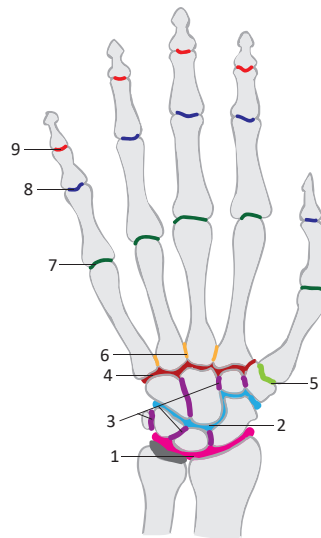
Anterior and posterior views of the right hand



The joints of the hand are a **group of joints** that work together to make complex hand movements possible.

Joints of the hand

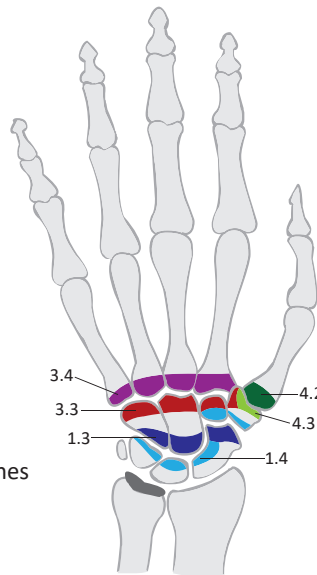
- **1 Wrist joint**
(*articulatio radiocarpalis*)
- **2 Midcarpal joint**
(*articulatio mediocarpalis*)
- **3 Intercarpal joints**
(*articulationes intercarpales*)
- **4 Carpometacarpal joints**
(*articulationes carpometacarpales*)
- **5 Carpometacarpal joint of thumb**
(*articulatio carpometacarpalis pollicis*)
- **6 Intermetacarpal joints**
(*articulationes intermetacarpales*)
- **7 Metacarpophalangeal joints**
(*articulationes metacarpophalangeae*)
- **8 Proximal interphalangeal joints**
(*articulationes interphalangeae proximales*)
- **9 Distal interphalangeal joints**
(*articulationes interphalangeae distales*)



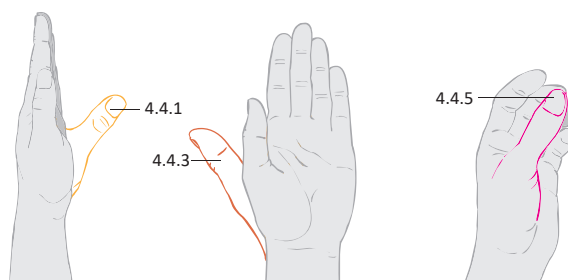
Anterior view of the right hand

Proximal joints of the hand

- 1 Midcarpal joint** (*articulatio mediocarpalis*)
 - 1.1 **Type:** compound
 - 1.2 **Shape:** of the letter S
 - 1.3 **Articular head:**
 - medially:** capitate and hamate
 - laterally:** scaphoid
 - 1.4 **Articular fossa:**
 - medially:** scaphoid, lunate and triquetrum
 - laterally:** trapezium and trapezoid
- 2 Intercarpal joints** (*articulationes intercarpales*)
 - joints in between the individual carpal bones
- 3 Carpometacarpal joints** (*articulationes carpometacarpales*)
 - 3.1 **Type:** compound
 - 3.2 **Shape:** plane
 - 3.3 **Articular head:** the distal line of the carpal bones
 - 3.4 **Articular fossa:** the base of the 2nd–5th metacarpal bones
- 4 Carpometacarpal joint of thumb**
(*articulatio carpometacarpalis pollicis*)
 - 4.1 **Shape:** saddle
 - 4.2 **Head:** the base of the 1st metacarpal bone
 - 4.3 **Cavity:** trapezium
 - 4.4 **Movements:**
 - 4.4.1 **Flexion:** 0–45°
 - 4.4.2 **Extension:** 0–15°
 - 4.4.3 **Abduction:** 70°
 - 4.4.4 **Adduction**
 - 4.4.5 **Opposition**
 - 4.4.6 **Reposition**
- 5 Intermetacarpal joints**
(*articulationes intermetacarpales*)
 - joints between the adjacent surface of the bases of the 2nd–5th metacarpal bones



Anterior view of the right hand



Some of the carpal ligaments on the palmar side of hand form bands in a shape of the letter V. Between them, there is a weak spot called the space of Poirier. The lunate bone can luxate through this space. The proximal part of the V band is formed by the radiotriquetral and ulnotriquetral ligament. The distal part of the V band is formed by the radiocapital and capitaltriquetral ligament.

Similar bands are found on the dorsum of the hand. These are the **radiotriquetral ligament**, which crosses the lunate bone and the smaller **triquetroscaphoid ligament**.

The **transverse carpal ligament** (*ligamentum carpi transversum*) is an obsolete term for flexor retinaculum.

The **pisiform joint** (*articulatio ossis pisiformis*) is a simple flat rigid joint. Its articular head is formed by the triquetrum and its socket by the pisiform. Ligaments stabilising the joint are the pisohamate and pisolmetacarpal ligament.

Palmar fibrocartilage (*fibrocartilago palmaris*) enlarges the articular fossa of metacarpophalangeal and interphalangeal joints. It is responsible for strengthening the articular capsule from the palmar side. Palmar fibrocartilage is located inside the **palmar ligaments** (*ligamenta palmaria*).

Clinical notes

Bouchard's nodes are fusiform swellings of the metacarpophalangeal and proximal interphalangeal joints that occur in rheumatoid arthritis. The distal interphalangeal joints are not usually affected by this process. Larger joints are usually affected in the later phase of the disease

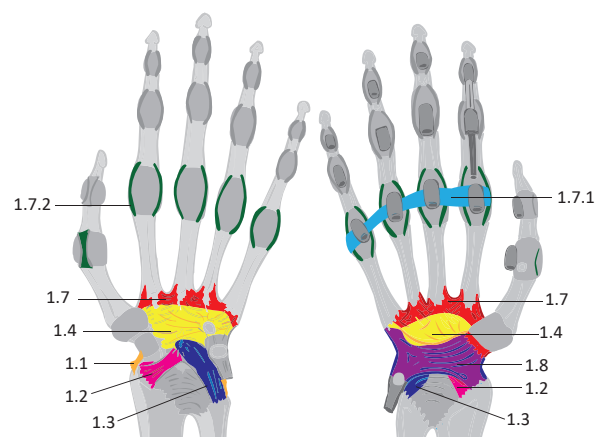
Heberden's nodes are thickenings of the distal interphalangeal joints caused by the development of bone spurs (osteophytes). These joint deformations are typical signs of osteoarthritis.

Joints of the hand as a group

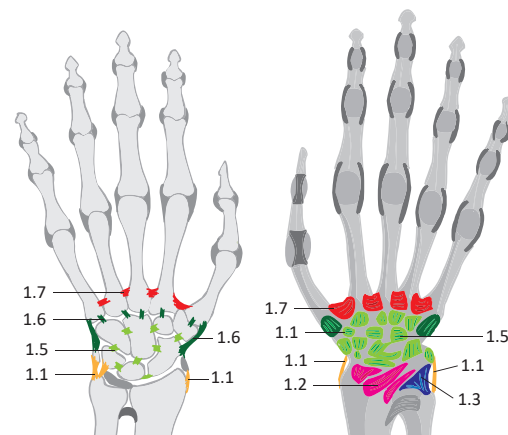
1 Ligaments: attach to all the bones that they pass

- 1.1 **Collateral ligaments of wrist joint** (*ligamenta collateralia carpi*)
 - weak ligaments that run on both sides of the wrist joint
- 1.2 **Palmar and dorsal ulnocarpal ligaments** (*ligamentum ulnocarpale palmare et dorsale*)
 - run from the ulna to the carpal bones
- 1.3 **Palmar and dorsal radiocarpal ligament** (*ligamentum radiocarpale palmare et dorsale*)
 - run from the radius to the carpal bones
- 1.4 **Radiate carpal ligament** (*ligamentum carpi radiatum*)
 - radiates outward from the palmar side of the capitate
- 1.5 **Interosseous intercarpal ligaments** (*ligamenta intercarpalia interossea*)
- 1.6 **Palmar and dorsal carpometacarpal ligaments** (*ligamenta carpometacarpalia palmaria et dorsalia*)
- 1.7 **Dorsal, palmar and interosseous metacarpal ligaments** (*ligamenta metacarpalia palmaria, dorsalia et interossea*)
 - 1.7.1 **Deep transverse metacarpal ligament** (*ligamentum metacarpale transversum profundum*)
 - 1.7.2 **Medial and lateral collateral ligament** (*ligamentum collaterale mediale et laterale*)
- 1.8 **Flexor retinaculum** (*retinaculum musculorum flexorum*)
 - a strong ligament stretched between the radial and ulnar carpal eminences
 - forms the roof of the carpal tunnel

2 Movements: very limited



Anterior view of the left and right hands



Right hand, viewed from the front and back

Joints of the fingers

1 Metacarpophalangeal joint (*articulatio metacarpophalangea*)

- 1.1 **Type:** simple
- 1.2 **Shape:** the palmar side of the joint is cylindrical-shaped but the dorsal side resembles a ball-and-socket joint (functionally ellipsoid joints)
- 1.3 **Articular head:** the head of the metacarpal
- 1.4 **Articular fossa:** the base of the proximal phalanx
- 1.5 **Ligaments:**
 - 1.5.1 **Medial and lateral collateral ligament** (*ligamentum collaterale mediale et laterale*)
 - 1.5.2 **Palmar ligaments** (*ligamenta palmaria*)
 - 1.5.3 **Deep transverse metacarpal ligament** (*ligamentum metacarpale transversum profundum*)

1.6 Accessory features:

- 1.6.1 **Palmar fibrocartilage** (*fibrocartilago palmaris*)

1.7 Movements:

- 1.7.1 **Flexion:** 90°
- 1.7.2 **Extension:** 0–20°
- 1.7.3 **Radial and ulnar duction** is possible only in extended joints
 - abduction away from the 3rd finger: 25–40°
 - adduction to wards the 3rd finger: 0–5°

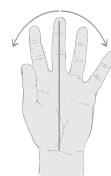
1.7.5 Passive rotations:

- 1.8 **Neutral position:** 45° of flexion of the fingers

2 Proximal and distal interphalangeal joints

(*articulationes interphalangeae proximales et distales*)

- 2.1 **Type:** simple, mobile
- 2.2 **Shape:** trochlear
- 2.3 **Head:** the head of the proximal and middle phalanges
- 2.4 **Cavity:** the base of the middle and distal phalanges
- 2.5 **Ligaments:**
 - 2.5.1 **Medial and lateral collateral ligament** (*ligamentum collaterale mediale et laterale*)
 - 2.5.2 **Palmar ligaments** (*ligamenta palmaria*)
- 2.6 **Accessories:**
 - 2.6.1 **Palmar fibrocartilage** (*fibrocartilago palmaris*)
- 2.7 **Movements:** flexion and extension from 70 to 100°
- 2.8 **Neutral position:** half the way to maximal flexion



Opening of fingers
– abduction from the 3rd finger



Closing of fingers
– adduction to the 3rd finger



Extension in the metacarpophalangeal
and interphalangeal joints



Flexion in the metacarpophalangeal
joints and extension
in the interphalangeal joints



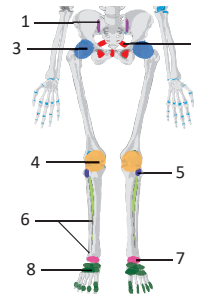
Extension in the metacarpophalangeal joints
and flexion in the proximal and distal
interphalangeal joints



Flexion in the metacarpophalangeal
and interphalangeal joints

The bones of the lower limb are connected predominantly by synovial joints. The bones of the leg, however, are connected to each other by an **interosseous membrane** and the **tibiofibular syndesmosis**.

- 1 Sacro-iliac joint (*articulatio sacroiliaca*)
- 2 Synarthroses of pelvic girdle (*synarthroses cinguli pelvici*)
- 3 Hip joint (*articulatio coxae*)
- 4 Knee joint (*articulatio genus*)
- 5 Tibiofibular joint / superior tibiofibular joint (*articulatio tibiofibularis*)
- 6 Interosseous membrane of leg and tibiofibular syndesmosis / inferior tibiofibular joint (*membrana interossea cruris et syndesmosis tibiofibularis*)
- 7 Ankle joint (*articulatio talocruralis*)
- 8 Joints of foot (*articulationes pedis*)



The **inferior pubic ligament** (*ligamentum pubicum inferius*), previously known as the arcuate ligament (*ligamentum arcuatum pubis*), is so strong that it can hold the two pubic bones together in the absence of the pubic symphysis.

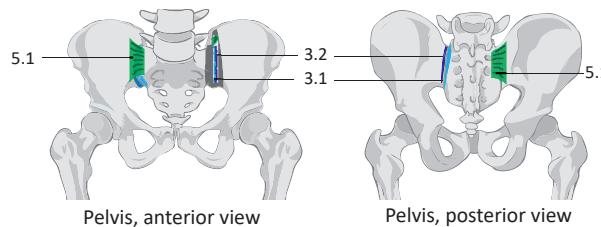
The **trochanteric fossa** is located outside of the joint capsule.

The **iliofemoral ligament** is short at birth but gains length during life. It is the strongest ligament in the human body. In newborns it keeps the thighs in a typical position (flexion, abduction and external rotation).

6.1 Sacro-iliac joint – Articulatio sacroiliaca

The **sacroiliac joint** is an amphiarthrosis between the sacrum and ilium that allows only very limited movements. It establishes the **angle of inclination of the pelvis** and transmits force between the pelvis and vertebral column.

- 1 **Type:** simple
- 2 **Shape:** plane
- 3 **Articular surfaces:**
 - 3.1 Auricular surface of sacrum (*facies auricularis ossis sacri*)
 - 3.2 Auricular surface of ilium (*facies auricularis ossis ilii*)
- 4 **Capsule:** short and firm
- 5 **Ligaments:**
 - 5.1 Anterior, posterior and interosseous sacro-iliac ligament (*ligamentum sacroiliacum anterius, posterius et interosseum*)
- 6 **Movements:** limited translational movements



Clinical notes

Overloading of the sacroiliac joint causes local pain that radiates into the thigh. This pain is a consequence of muscular spasms that attempt to stabilise the overloaded joint. The most frequent causes of sacroiliac joint overload are certain sports, working in a forward bended position and osteoarthrotic changes.

Pain over the pubic symphysis can be caused by hypertonic hip joint adductors.

Dislocation of the hip joint may be caused by a high energy trauma. This typically occurs in car accidents when the knees are impacted on the car dashboard (the so called dashboard injury). 90 % of the cases are posterior dislocations, which occur through a weak spot between the ligaments strengthening the articular capsule.

Developmental dysplasia of the hip is a congenital deformity of the hip joint which, if not treated, causes orthopedic problems throughout life. It is characterised by a misshapen acetabulum and/or head and neck of the femur. This causes a misalignment of the "bone-to-bone" contact between the articular head and fossa. All newborns are screened for DDH after birth with a specialised examination that includes ultrasound.

Total hip replacement (endoprosthesis) is an orthopaedic procedure that replaces the articular head and fossa of the hip joint with an artificial prosthesis. Severely damaged hips are indicated for this operation, such as occur in osteoarthrosis or after injuries.

Hemiarthraphy of the hip is a replacement of the articular head only. This is usually indicated in femoral neck fractures occurring in elderly people.

6.2 Synarthroses of the pelvic girdle – Synarthroses cinguli pelvici

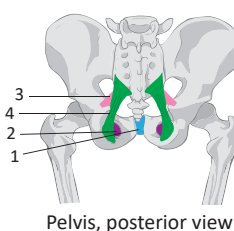
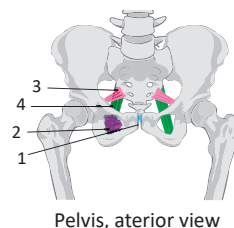
The **pubic symphysis** is an elastic cartilaginous joint uniting the two pubic bones in the midline. A fluid-filled sagittal fissure can develop within the cartilaginous disc, imparting a resemblance of a diarthrosis. The hormonal changes occurring during pregnancy loosen the cartilage, preparing the pelvis for parturition.

Pubic symphysis (*symphysis pubica*)

- 1 **Pubic symphysis**
 - 1.1 **Type of connection:** symphysis
 - 1.2 **Accessory features:** the symphyseal surfaces of the pubic bones are covered by hyaline cartilage which is continuous with a fibrocartilaginous articular disc
 - 1.3 **Ligaments:** the superior and inferior pubic ligaments (*ligamentum pubicum superius et inferius*)

Syndesmoses of the pelvic girdle (*syndesmoses cinguli pelvici*)

- 2 **Obturator membrane** (*membrana obturatoria*)
 - covers the obturator foramen
 - contains a small opening, the obturator canal (*canalis obturatorius*), for the obturator vessels and nerve
- 3 **Sacrospinous ligament** (*ligamentum sacrospinale*)
 - runs from the ischial spine to the sacrum and coccyx
 - closes the greater sciatic notch forming the greater sciatic foramen
- 4 **Sacrospinous ligament** (*ligamentum sacrotuberale*)
 - runs from the ischial tuberosity to the sacrum and coccyx
 - closes the lesser sciatic notch forming the lesser sciatic foramen



The **hip joint** is a **limited synovial ball-and-socket joint** (a cotyloid joint) which allows **limited movements in all three planes**. The deep articular fossa is formed by the **acetabulum of the pelvis** which articulates with the **head of the femur**. The articular surface of the acetabulum is called the **lunate articular facet** and is enlarged by a **cartilaginous labrum**. The articular capsule is thickened by **three strong ligaments** that run from the pubis, ilium and ischium to the femur.

1 **Type:** simple

2 **Shape:** limited ball-and-socket (cotyloid)

3 **Articular head:** the head of the femur

4 **Articular fossa:** acetabulum

4.1 **Lunate surface** (*facies lunata*) – the articular facet in the acetabulum

5 **Capsule:** attaches to the margins of the acetabulum and across the posterior aspect of the neck (1 cm above the intertrochanteric crest)

5.1 **Retinacula of Weitbrecht** – folds of a synovial membrane formed by impressions of the vessels supplying the neck and head of the femur

6 **Ligaments:**

6.1 **Iliofemoral ligament** (*ligamentum iliofemorale*)

– prevents hyperextension of the hip joint

6.2 **Pubofemoral ligament** (*ligamentum pubofemorale*)

– limits abduction and external rotation

6.3 **Ischiofemoral ligament** (*ligamentum ischiofemorale*)

– limits adduction and internal rotation

6.3.1 **Zona orbicularis**

– a ligamentous ring encircling the neck of the femur formed by the pubofemoral and ischiofemoral ligaments

6.4 **Transverse acetabular ligament**

(*ligamentum transversum acetabuli*)

– bridges the acetabular notch

6.5 **Ligament of head of femur** (*ligamentum capitis femoris*)

– a small ligament connecting the transverse acetabular ligament and fovea for the ligament of the head of the femur

– contains the artery of the head of the femur

7 **Accessory features:**

7.1 **Acetabular labrum** (*labrum acetabuli*) – a fibrocartilaginous rim that enlarges the cavity of the acetabulum

7.2 **Acetabular fat pad** (*pulvinar acetabuli*) – a fat pad in the acetabular fossa

8 **Movements:**

8.1 **Flexion:** 0–130°

8.2 **Extension:** 0–30°

– limited by tension of the iliofemoral ligament

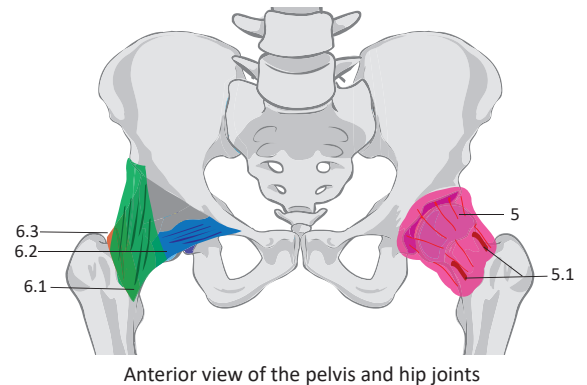
8.3 **Abduction:** 0–45°

8.4 **Adduction** (hyperadduction): 0–30°

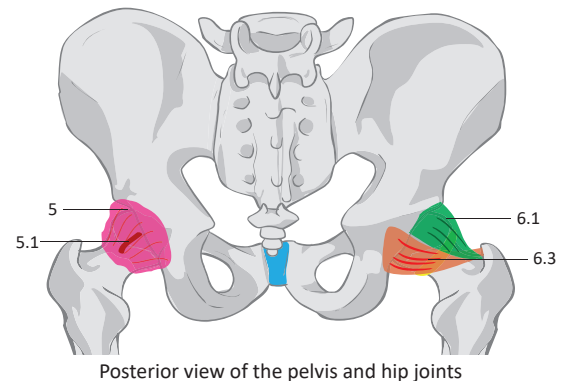
8.5 **External rotation:** to 45°

8.6 **Internal rotation:** to 35°

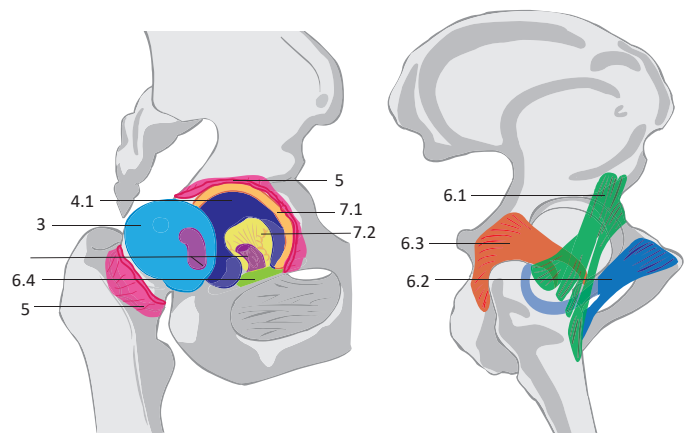
9 **Neutral position:** mild flexion, mild external rotation and mild abduction



Anterior view of the pelvis and hip joints

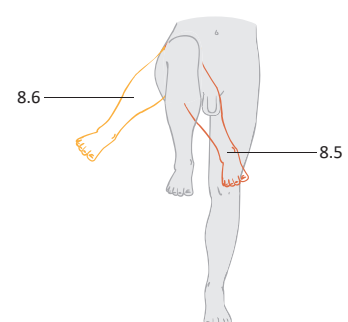
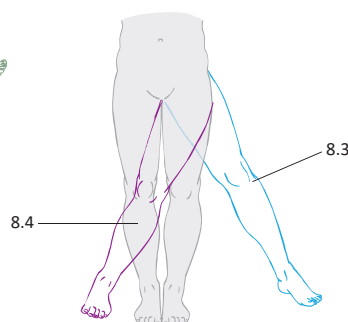
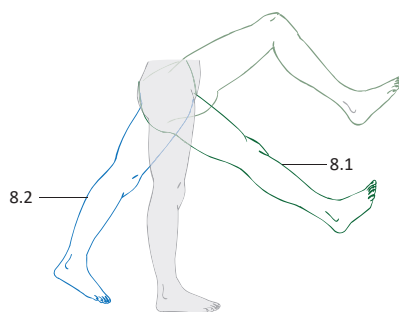


Posterior view of the pelvis and hip joints



Open right hip joint viewed from the right

Right hip joint viewed from the right



The **knee joint** is the most complicated synovial joint in the human body. It is a **complex and compound bicondylar joint** in which **three bones and two menisci** articulate. The knee joint has **12 articular ligaments** and many **associated bursae**. Movements are possible in **two axes**, but these movements are limited when the knee is in full flexion and full extension.

Type, shape and articular surfaces of the knee joint

Type: both complex and compound

1 Femoropatellar joint (*articulatio femoropatellaris*)

1.1 Shape: plane

- 1.2 **Articular head:** the patellar surface of the femur
- 1.3 **Articular fossa:** the articular surface of the patella

2 Femorotibial joint (*articulatio femorotibialis*)

2.1 Shape: a combination of the hinge and pivot joints

- 2.2 **Articular head:** the condyles of the femur
- 2.3 **Articular fossa:** the condyles of the tibia

Articular capsule

1 Fibrous capsule

- originates below the epicondyles of the femur
- attaches to the circumference of the articular surfaces of the tibia and patella

2 Synovial membrane

- lines the internal surface of the fibrous layer
- does not cover the cruciate ligaments

2.1 Infrapatellar synovial fold

(*plica synovialis infrapatellaris*)

- 2.1.1 **Alar folds** (*plicae alares*) – project dorsally from the ventral part of the joint

2.2 Infrapatellar fat pad

(*corpus adiposum infrapatellare*)

- a fat pad between the fibrous and synovial layers

3 Synovial bursae and articular recesses

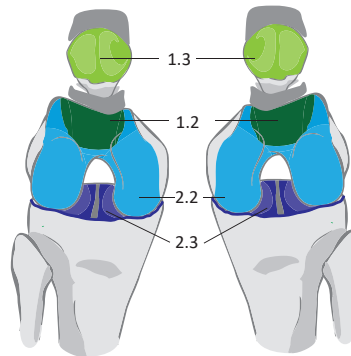
- approximately 20 bursae and 2 recesses are located in the vicinity of the knee joint

Communicating with the knee joint:

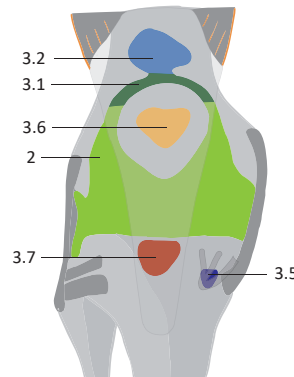
- 3.1 **Suprapatellar recess** (*recessus suprapatellaris*)
- 3.2 **Suprapatellar bursa** (*bursa suprapatellaris*)
- 3.3 **Subpopliteal recess** (*recessus subpopliteus*)
- 3.4 **Gastrocnemiosemimembranosus bursa** (*bursa gastrocnemiosemimembranosa*)

Not communicating with the knee joint:

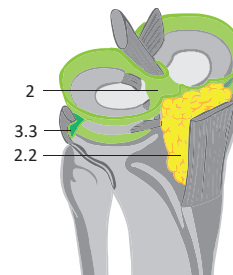
- 3.5 **Anserine bursa** (*bursa anserina*)
- 3.6 **Subcutaneous prepatellar bursa** (*bursa subcutanea prepatellaris*)
- 3.7 **Subcutaneous infrapatellar bursa** (*bursa subcutanea infrapatellaris*)



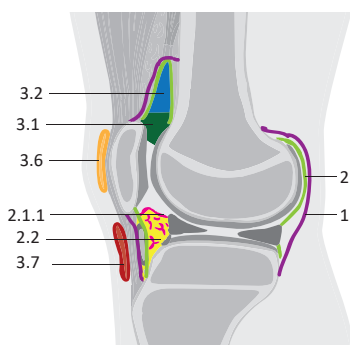
Open right and left knee joints



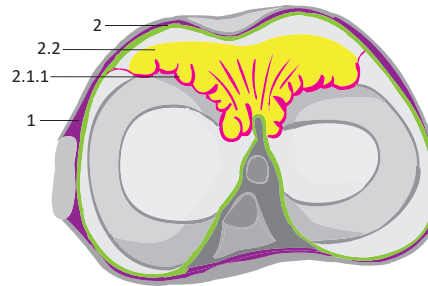
Anterior view of the right knee joint



Right anterior superior view of the right knee joint



Sagittal section of the right knee joint viewed from the left



Superior view of a transverse section of the right knee joint

The **valve mechanism** of the knee is a system of connected pouches and synovial bursae of the knee joint through which synovial fluid flows. The ventral part is formed by the suprapatellar recess and bursa, the dorsomedial part is formed by the gastrocnemius-semimembranosus bursa and the dorsolateral part by the subpopliteal recess.

A mnemonics for remembering the positions of the cruciate ligaments: when the middle finger is crossed over the index finger and the hand placed on the ipsilateral knee, the middle finger shows the position of the anterior cruciate ligament and the index finger shows the position of the posterior cruciate ligament.

In clinical practice, the tibial and fibular collateral ligaments are inaccurately termed the medial and lateral collateral ligaments.

The middle position of the knee joint is 20–30° of flexion.

The articularis genus is a stand-alone muscle below the quadriceps femoris. It attaches to the joint capsule, which stretches up and prevents its incarceration.

Clinical notes

The unhappy triad is a term given to concomitant damage to the anterior cruciate ligament, medial meniscus and tibial collateral ligament.

The menisci, anterior cruciate ligament and collateral ligaments are the most predisposed structures of the knee to be injured. The anterior cruciate ligament is injured ten times more frequently than the posterior cruciate ligament.

Baker's cyst arises when the gastrocnemius-semimembranosus bursa becomes overfilled with synovial fluid. It is palpable in the subcutaneous tissue of the popliteal fossa.

Total knee replacement is a surgical procedure that replaces severely damaged femoral and tibial condyles with artificial prostheses. This is performed when conservative therapy with analgetics, injections, and rehabilitation is not sufficient in patients with severe arthrosis. Other indications include disease of the knee caused by systemic diseases such as rheumatoid arthritis, ankylosing spondylitis disease and psoriatic arthritis.

Ligaments

1 Collateral ligaments:

- 1.1 **Tibial collateral ligament** (*ligamentum collaterale tibiale*)
 - runs from the medial epicondyle of the femur to the tibia
 - is attached firmly to the medial meniscus
- 1.2 **Fibular collateral ligament** (*ligamentum collaterale fibulare*)
 - runs from the lateral epicondyle of the femur to the head of the fibula

2 Ventral ligaments:

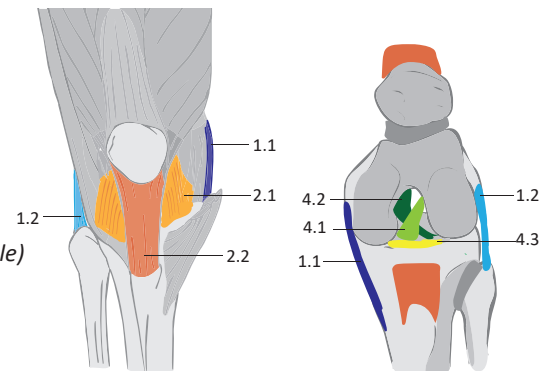
- 2.1 **Medial and lateral patellar retinacula** (*retinaculum patellae mediale et laterale*)
 - attach the quadriceps femoris directly to the tibial tuberosity on either side of the patella
 - hold the patella in the shallow patellar surface of the femur
- 2.2 **Patellar ligament** (*ligamentum patellae*)
 - the insertion of the quadriceps femoris between the patella and the tibial tuberosity

3 Dorsal ligaments:

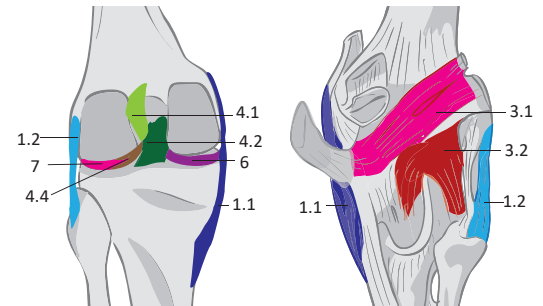
- 3.1 **Oblique popliteal ligament** (*ligamentum popliteum obliquum*)
 - one of the insertions of the semimembranosus
 - is also known as the pes anserinus profundus
- 3.2 **Arcuate popliteal ligament** (*ligamentum popliteum arcuatum*)
 - forms an arch above the tendon of the popliteus

4 Intra-articular ligaments:

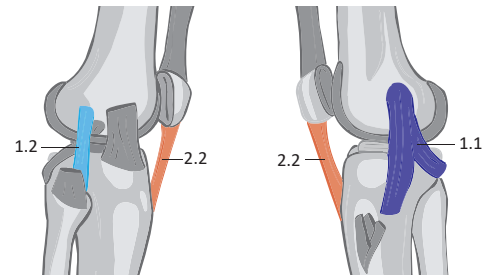
- 4.1 **Anterior cruciate ligament** (*ligamentum cruciatum anterius*)
 - runs from the medial surface of the lateral condyle of the femur ventromedially to the anterior intercondylar area of the tibia
- 4.2 **Posterior cruciate ligament** (*ligamentum cruciatum posterius*)
 - runs from the lateral surface of medial condyle of the femur caudally to the posterior intercondylar area of the tibia
- 4.3 **Transverse ligament of knee** (*ligamentum transversum genus*)
 - connects both menisci ventrally; not covered by synovial membrane (the only true intra-articular ligament of the human body)
- 4.4 **Anterior and posterior menisiofemoral ligament** (*ligamentum menisiofemorale anterius et posterius*)
 - weak ligaments around the posterior cruciate ligament



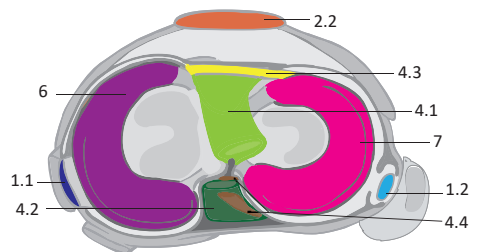
Posterior view of the right knee in extension and the left knee in flexion with the patella reflected



Posterior view of the left and right knee joints



Medial and lateral sides of the right knee joint



Superior view of the right knee joint

Menisci

- 6 **Medial meniscus** (*meniscus medialis*) – a C-shape semicircle
 - attaches to the anterior and posterior intercondylar area
 - is firmly attached medially to the articular capsule and laterally to the tibial collateral ligament, which limits its mobility
 - the semimembranosus prevents displacement by pulling it at its dorsal end
- 7 **Lateral meniscus** (*meniscus lateralis*) – almost forms a complete circle
 - attaches to the anterior and posterior intercondylar area
 - is not attached to the capsule and so is more mobile than the medial meniscus
 - the popliteus prevents displacement by pulling its dorsal end

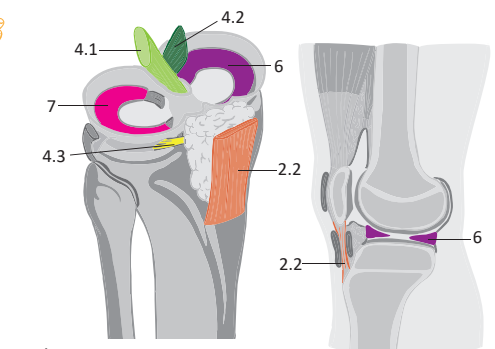
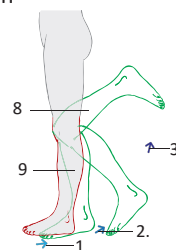
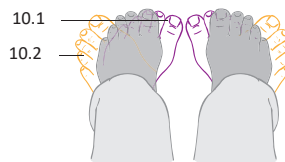
Movements

- 8 **Flexion:** 0–150°
- 9 **Extension:** 0°
- 10 **Rotations** – is possible only in flexed knees
 - 10.1 **Internal rotation:** 5–10°
 - 10.2 **External rotation:** approximately 40°

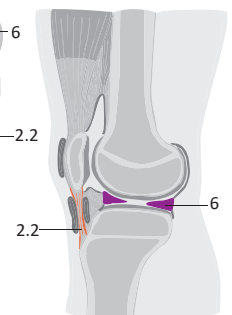
Flexion of the knee joint (3 phases):

– the patella slips distally during flexion and proximally during extension

1. **Initial rotation** (unlocking of knee joint): flexion to 5°
 - release of the cruciate ligaments and tibial collateral ligament
 - tibia rotates internally
 - lockout of the knee occurs in the reverse order
2. **Rolling movement:** flexion 10–20°
 - condyles of the femur roll over the condyles of the tibia and on the menisci
3. **Sliding movement:** from 20° flexion to approximately 160°
 - the condyles of the femur and menisci slide on the tibia dorsally



Right anterior superior view of the right knee joint

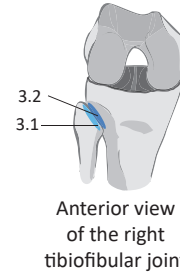


Sagittal section of the left knee joint, view from the right

6.5 Tibiofibular joint (superior tibiofibular joint) – Articulatio tibiofibularis

The tibiofibular joint is a plane synovial joint that allows only very limited movement. It is an amphiarthrosis.

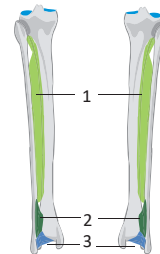
- 1 **Type:** simple
- 2 **Shape:** plane
- 3 **Articular surfaces:**
 - 3.1 **Articular facet of head of fibula** (*facies articularis capitis fibulae*)
 - 3.2 **Fibular articular facet of tibia** (*facies articularis fibularis tibiae*)
- 4 **Capsule:** firm and small
- 5 **Ligaments:** the anterior and posterior ligaments of the head of the fibula (*ligamentum capitis fibulae anterius et posterius*)
- 6 **Movements:** passive movements due to the movements of the ankle joint



6.6 Interosseous membrane of the leg and tibiofibular syndesmosis

The interosseous membrane of the leg runs from the tibia laterodistally and prevents the tibia and fibula moving away from each other. The muscles of the deep layer of the leg originate on this membrane. **The tibiofibular syndesmosis** connects the distal part of the tibia and fibula, which forms a fork shaped unit for articulation with the talus.

- 1 **Interosseous membrane of leg** (*membrana interossea cruris*)
 - attaches to the interosseous borders of the fibula and tibia
- 2 **Tibiofibular syndesmosis / inferior tibiofibular joint** (*syndesmosis tibiofibularis*)
 - a distal thickening of the interosseous membrane
- 3 **Anterior and posterior tibiofibular ligaments** (*ligamentum tibiofibulare anterius et posterius*)
 - ventral and dorsal strengthenings of the tibiofibular syndesmosis



6.7 Joints of the foot – Articulationes pedis

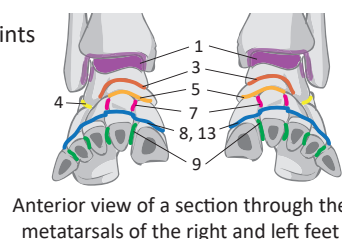
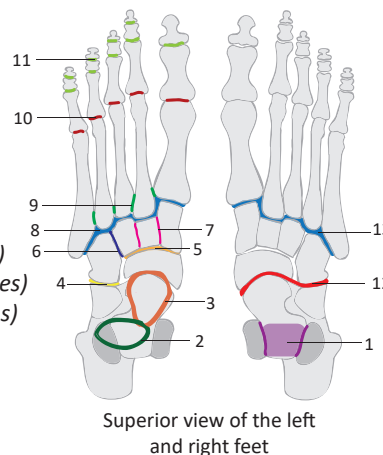
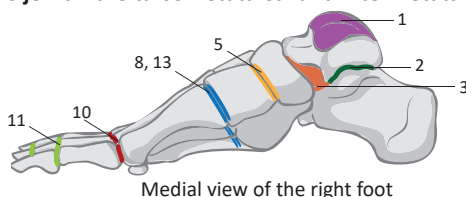
The joints of the foot comprise a functional unit that produces foot movements necessary for ambulation.

Joints of the foot

- 1 **Ankle joint** (*articulatio talocruralis*)
- 2 **Subtalar/talocalcaneal joint** (*articulatio subtalaris/talocalcanea*)
- 3 **Talocalcaneonavicular joint** (*articulatio talocalcaneonavicularis*)
- 4 **Calcaneocuboid joint** (*articulatio calcaneocuboidea*)
- 5 **Cuneonavicular joint** (*articulatio cuneonavicularis*)
- 6 **Cuneocuboid joint** (*articulatio cuneocuboidea*)
- 7 **Intercuneiform joints** (*articulationes intercuneiformes*)
- 8 **Tarsometatarsal joints** (*articulationes tarsometatarsales*)
- 9 **Intermetatarsal joints** (*articulationes intermetatarsales*)
- 10 **Metatarsophalangeal joints** (*articulationes metatarsophalangeae*)
- 11 **Interphalangeal joints of foot** (*articulationes interphalangeae pedis*)

Functional joints

- 12 **Chopart's joint** – the transverse tarsal joint
- 13 **Lisfranc's joint** – the tarsometatarsal and intermetatarsal joints



Joints

All movements in the tarsus are associated with rotation of the fibula

The articular facet of the medial malleolus is lunate or tear-drop shaped.

The articular facet of the lateral malleolus has a triangular shape.

The opening for the anterior tibial vessels (*vasa tibialia anteriora*) is located at the proximal end of the interosseous membrane of the leg.

The bifurcate ligament is a strong ligament on the dorsum of the foot. Distal to its attachment on the calcaneus, it bifurcates into the calcaneocuboid and calcaneonavicular ligaments.

The spring ligament is another name for the plantar calcaneonavicular ligament.

The plantar calcaneocuboid ligament is also known as the short plantar ligament.

Talotarsal joint (subtalar/talocalcaneal and talocalcaneonavicular joints) are a functional unit for the inversion and eversion.

Clinical notes

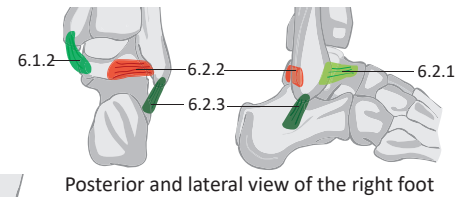
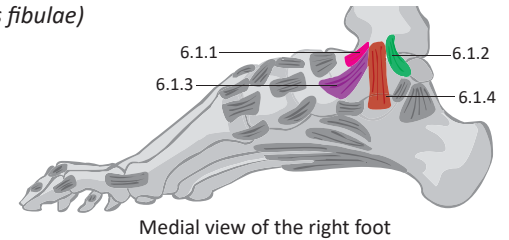
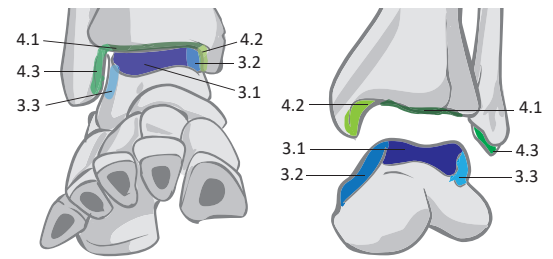
The tibiofibular syndesmosis is a very firm synarthrosis. A trauma to the leg is much more likely to cause dislocation of the ankle than rupture of the tibiofibular syndesmosis.

Ankle fractures are classified preferably according to Weber into types A, B and C. This classification describes the fractures according to the location of fracture in relation to the tibiofibular syndesmosis.

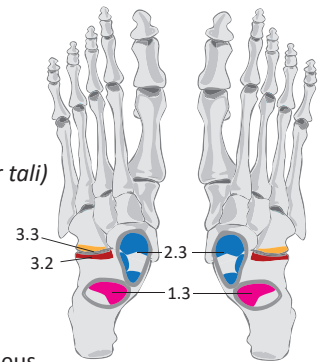
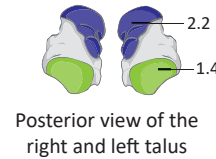
Injuries of the collateral ligaments of the ankle represent some of the most frequent types of sport injuries. The medial collateral ligament is firmer than the lateral and so is injured less often. A sprained ankle, the most common type of ankle injury, is caused by a tear in the anterior inferior talofibular ligament.

Ankle joint (*articulatio talocruralis*)

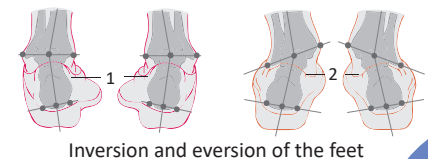
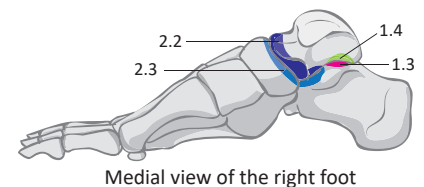
- 1 **Type:** a compound joint of three articulating bones
- 2 **Shape:** trochlear
- 3 **Articular head:** the trochlea of the talus
 - 3.1 **Superior facet** (*facies articularis superior*)
 - 3.2 **Medial malleolar facet** (*facies articularis malleolaris medialis tali*)
 - 3.3 **Lateral malleolar facet** (*facies articularis malleolaris lateralis tali*)
- 4 **Articular fossa:** the distal end of the tibia and fibula
 - 4.1 **Inferior articular surface of tibia** (*facies articularis inferior tibiae*)
 - 4.2 **Articular facet of medial malleolus of tibia** (*facies articularis malleoli medialis tibiae*)
 - 4.3 **Articular facet of lateral malleolus of fibula** (*facies articularis malleoli lateralis fibulae*)
- 5 **Capsule:** is attached to the circumference of the articular surfaces
- 6 **Collateral ligaments:**
 - 6.1 **Medial/deltoid ligament** (*ligamentum collaterale mediale / ligamentum deltoideum*)
 - a triangular ligament composed of four parts
 - 6.1.1 **Anterior tibiotalar part** (*pars tibiotalaris anterior*)
 - 6.1.2 **Posterior tibiotalar part** (*pars tibiotalaris posterior*)
 - 6.1.3 **Tibionavicular part** (*pars tibionavicularis*)
 - 6.1.4 **Tibiocalcaneal part** (*pars tibiocalcanea*)
 - 6.2 **Lateral ligament** (*ligamentum collaterale laterale*)
 - a complex of three independent ligaments
 - 6.2.1 **Anterior talofibular ligament** (*ligamentum talofibulare anterius*)
 - 6.2.2 **Posterior talofibular ligament** (*ligamentum talofibulare posterius*)
 - 6.2.3 **Calcaneofibular ligament** (*ligamentum calcaneofibulare*)
- 7 **Movements:**
 - 7.1 **Plantar flexion:** 0–50°
 - 7.2 **Dorsiflexion:** 0–20°
- 8 **Middle position:** corresponds to the anatomical position

**Subtalar/talocalcaneal joint (*articulatio subtalaris/talocalcanea*)**

- 1.1 **Type:** simple
- 1.2 **Shape:** cylindrical
- 1.3 **Articular head:** the posterior articular talar surface of the calcaneus (*facies articularis talaris posterior calcanei*)
- 1.4 **Articular fossa:** the posterior calcaneal articular facet of the talus (*facies articularis calcanea posterior tali*)
- 1.5 **Ligaments:** the posterior, medial, lateral and interosseous talocalcaneal ligaments (*ligamentum talocalcaneum posterius, mediale, laterale et interosseum*)

**Talocalcaneonavicular joint (*articulatio talocalcaneonavicularis*)**

- 2.1 **Type:** a compound joint of three articulating bones
- 2.2 **Articular head:** the head of the talus and the anterior and middle articular talar surfaces of the calcaneus
- 2.3 **Articular fossa:** the navicular and the anterior and middle calcaneal articular facets of the talus and navicular fibrocartilage
- 2.4 **Ligaments:** the talonavicular ligament (*ligamentum talonaviculare*), plantar calcaneonavicular ligament (*ligamentum calcaneonaviculare plantare*), and dorsal calcaneonavicular ligament (*ligamentum calcaneonaviculare dorsale*)
- 2.5 **Accessory features:** the navicular fibrocartilage which enlarges the fossa for the head of the talus (within the plantar calcaneonavicular ligament)

**Calcaneocuboid joint (*articulatio calcaneocuboidea*)**

- 3.1 **Type:** amphiarthroses
- 3.2 **Articular head:** the articular surface for the cuboid on the calcaneus
- 3.3 **Articular fossa:** cuboid
- 3.4 **Ligaments:** the plantar calcaneocuboid ligament (*ligamentum calcaneocuboideum plantare*) and dorsal calcaneocuboid ligament (*ligamentum calcaneocuboideum dorsale*)
- 3.5 **Movements:** very limited

Movements of the foot

- 1 **Inversion:** 0–30°
- 2 **Eversion:** 0–15°

Other foot joints

1 **Cuneonavicular joint** (*articulatio cuneonavicularis*)

– a compound plane joint between the cuneiform bones and navicular

- 1.1 **Ligaments:** the plantar and dorsal cuneonavicular ligament (*ligamentum cuneonaviculare plantare et dorsale*)

2 **Cuneocuboid joint** (*articulatio cuneocuboidea*)

– a plane joint between the lateral cuneiform and the cuboid

- 2.1 **Ligaments:** the plantar, dorsal and interosseous cuneocuboid ligaments (*ligamentum cuneocuboideum plantare, dorsale et interosseum*) and plantar and dorsal cuboideonavicular ligaments (*ligamentum cuboideonaviculare plantare et dorsale*)

3 **Intercuneiform joints** (*articulationes intercuneiformes*)

– plane joints between the cuneiform bones

- 3.1 **Ligaments:** the plantar, dorsal and interosseous intercuneiform ligament (*ligamentum intercuneiforme plantare, dorsale et interosseum*)

4 **Tarsometatarsal joints** (*articulationes tarsometatarsales*)

4.1 **Type:** simple

4.2 **Shape:** plane

4.3 **Articular surfaces:**

- the distal surfaces of the cuneiform bones articulate with the bases of the 1st–3rd metatarsals
- the distal surface of the cuboid articulates with the bases of the 4th–5th metatarsals

- 4.4 **Ligaments:** the plantar and dorsal tarsometatarsal ligaments (*ligamenta tarsometatarsalia plantaria et dorsalia*) and interosseous cuneometatarsal ligaments (*ligament cuneometatarsalia interossea*)

5 **Intermetatarsal joints** (*articulationes intermetatarsales*)

– connect the adjacent surfaces of the bases of the 1st–5th metatarsal bones

- 5.1 **Ligaments:** the plantar, dorsal and interosseous metatarsal ligaments (*ligamenta metatarsalia plantaria, dorsalia et interossea*)

6 **Metatarsophalangeal joints** (*articulationes metatarsophalangeae*)

6.1 **Articular head:** the head of the metatarsal bones

6.2 **Articular fossa:** the base of the adjacent proximal phalanges

6.3 **Ligaments:**

- 6.3.1 **Plantar and collateral ligaments** (*ligamenta plantaria et collateralia*)
- 6.3.2 **Deep transverse metatarsal ligament** (*ligamentum metatarsale transversum profundum*)
- connects the adjacent heads of the metatarsals

6.4 **Accessory features:** plantar fibrocartilage (*fibrocartilago plantaris*)

6.5 **Movements:** flexion, extension and mild adduction and abduction

7 **Proximal and distal interphalangeal joints of foot**

(*articulationes interphalangeae pedis proximales et distales*)

7.1 **Head:** the head of the proximal and middle phalanx

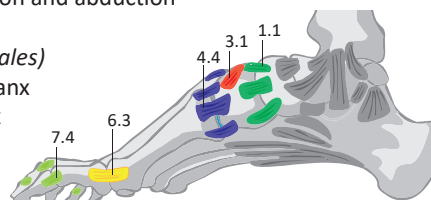
7.2 **Cavity:** the base of the middle and distal phalanx

7.3 **Shape:** trochlear

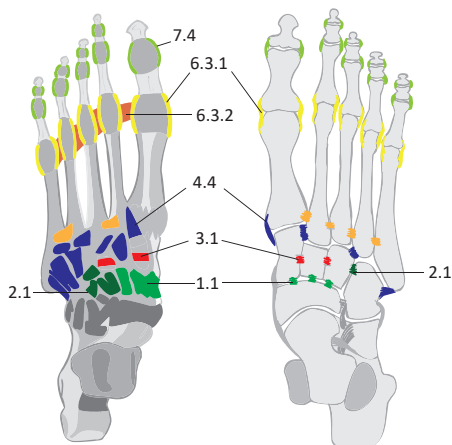
- 7.4 **Ligaments:** the collateral and plantar ligaments (*ligamenta collateralia et plantaria*)

7.5 **Accessories:** plantar fibrocartilage

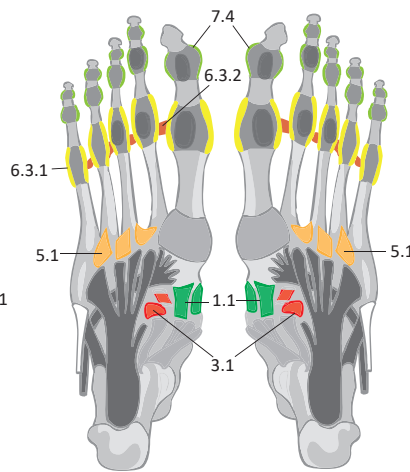
7.6 **Movements:** flexion and extension



Medial view of the right foot



Superior view of the right and left feet



Posterior view of the right and left feet

The key to **Chopart's joint** (*clavis articulationis tarsi transversae*) is a clinical term for the bifurcated ligament.

The **tendinous stirrup** is formed by the inserting tendons of tibialis anterior and fibularis longus at the medial cuneiform bone and supports the transverse arch of foot.

The **plantar fibrocartilage** (*fibrocartilago plantaris*) enlarges the articular heads of the metatarsophalangeal and interphalangeal joints, strengthening these joints. Plantar fibrocartilage is embedded within the **plantar ligaments** (*ligamenta plantaris*).

The **navicular bone** serves as a central keystone of the bony vault at the apex of the longitudinal arch of foot.

Clinical notes

Exarticulation of the distal part of the foot can be performed along the Chopart's or Lisfranc's joint lines.

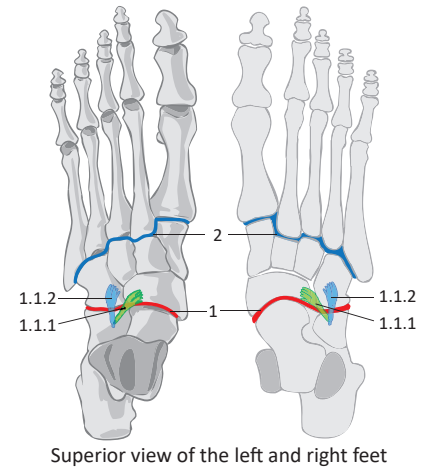
Hallux valgus is a deformity of the big toe characterised by a lateral deviation of the big toe towards the other toes. It can occur secondarily due to a varus deformity of the 1st metatarsal or to degenerative changes of the 1st metatarsophalangeal joint.

Pes transversoplanus is a type of flat foot caused by the collapse of the transverse arch of the foot.

Pes cavus is a foot deformity characterised by a high longitudinal arch of the foot.

Functional joints

- **1 Chopart's joint / transverse tarsal joint** (*articulatio tarsi transversa*)
 - is composed of the talonavicular joint, which is a part of the talocalcaneonavicular joint and the calcaneocuboid joint
 - a functionally and clinically important joint of the foot
 - allows the pronation and supination and enlarges the movement of the ankle joint
- 1.1 **Bifurcate ligament** (*ligamentum bifurcatum*)
 - is located on the dorsal aspect of the foot and composed of two parts
 - 1.1.1 **Calcaneonavicular ligament** (*ligamentum calcaneonaviculare*)
 - 1.1.2 **Calcaneocuboid ligament** (*ligamentum calcaneocuboideum*)
- **2 Lisfranc's joint**
 - comprises the tarsometatarsal and intermetatarsal joints
 - facilitates the spring movements of the foot
 - the serrate line of articulation is formed by the insertion of the 2nd metatarsal into a shallow cavity formed by the intermediate cuneiform, which is shorter than other cuneiform bones



Superior view of the left and right feet

6.7.1

Arches of the foot

The human foot is shaped by two longitudinal and one transverse arch. These arches are formed and maintained by the architecture of the tarsus and metatarsal bones, **the long and short ligaments of the foot and the muscles of the foot and leg.** The arches are fully developed by the 3rd year of life, at which point the foot is in contact with the ground in three spots: **the calcaneal tuberosity, the head of the 1st metatarsal bone and the head of the 5th metatarsal bone.** The arches of the foot **protect the soft tissue of the foot** and function as a **spring, converting elastic energy into kinetic energy during gait.**

- **1 Longitudinal arch** – two lines of bones which end on the navicular
 - **Medial:** the calcaneus, talus, navicular, three cuneiform bones and 1st metatarsal
 - 15–18 mm above the ground at its highest point
 - **Lateral:** the calcaneus, cuboid and 5th metatarsal
 - 3–5 mm above the ground at its highest point

Maintained by:

1.1 Longitudinal plantar ligaments

- 1.1.1 **Long plantar ligament** (*ligamentum plantare longum*)
 - a strong thick ligament that runs from the calcaneus to the bases of the metatarsals on the plantar aspect of the foot
 - has some fibres which run to the cuboid
- 1.1.2 short plantar and interosseous ligaments of the foot

1.2 Plantar aponeurosis (*aponeurosis plantaris*)

1.3 Muscles:

- 1.3.1 **Tibialis anterior** (*musculus tibialis anterior*)
- 1.3.2 **Tibialis posterior** (*musculus tibialis posterior*)
- 1.3.3 **Flexor hallucis longus** (*musculus flexor hallucis longus*)
- 1.3.4 **Flexor digitorum longus and brevis**
(*musculus flexor digitorum longus et brevis*)
- 1.3.5 **Intrinsic muscles of the foot**

● 2 Transverse arch:

- runs over the heads of all the metatarsals
- most developed in over the cuneiform bones and the cuboid
- held by the tendinous strirrup of the tibialis anterior and fibularis longus

Maintained by:

2.1 Transverse ligaments of the foot:

2.1.1 Plantar and interosseous intercuneiform ligaments

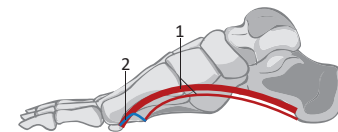
(*ligamenta intercuneiformia plantaria et interossea*)

2.1.2 Plantar cuneonavicular ligaments (*ligamenta cuneonavicularia plantaria*)

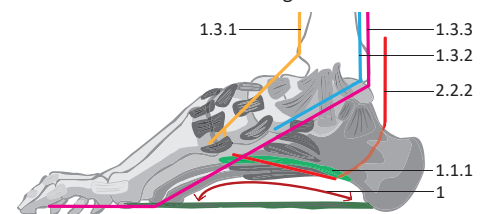
2.1.3 Plantar tarsometatarsal ligaments (*ligamenta tarsometatarsalia plantaria*)

2.2 Muscles:

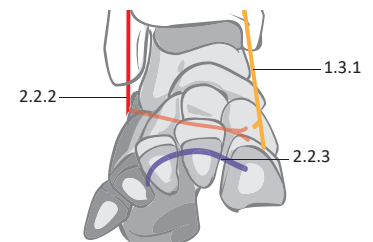
- 2.2.1 **Tibialis anterior** (*musculus tibialis anterior*)
- 2.2.2 **Fibularis longus** (*musculus fibularis longus*)
- 2.2.3 **Adductor hallucis** (*musculus adductor hallucis*)
- 2.2.4 **Intrinsic muscles of the foot**



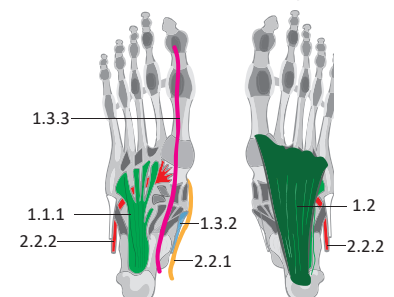
Medial view of the right foot



Medial view of the right foot



Anterior view of a section through the metatarsals in the right foot



Inferior view of the right and left feet

Head and neck			
Flexion / Ventral flexion (bilateral contraction)	Extension / Retroflexion / Dorsiflexion (bilateral contraction)	Lateral flexion / Lateroflexion / Inclination (unilateral contraction)	Rotation (unilateral contraction)
Sternocleidomastoid	Sternocleidomastoid	Sternocleidomastoid	Sternocleidomastoid
Scalenus anterior	Trapezius (with a fixed shoulder girdle)	Trapezius	Splenius capitis
Scalenus medius	Splenius capitis	Levator scapulae (unilateral contraction with a fixed scapula)	Splenius cervicis
Scalenus posterior	Splenius cervicis	Splenius capitis	Semispinalis capitis
Longus capitis	Longissimus capitis	Splenius cervicis	Semispinalis cervicis
Longus colli	Longissimus cervicis	Longissimus capitis	Longissimus capitis
Rectus capitis anterior	Semispinalis capitis	Longissimus cervicis	Longissimus cervicis
Suprahyoid and infrahyoid muscles working together synergistically	Semispinalis cervicis	Semispinalis capitis	Obliquus capitis inferior
	Rectus capitis posterior major	Semispinalis cervicis	Scalenus anterior
	Rectus capitis posterior minor	Rectus capitis posterior major	Scalenus medius
	Obliquus capitis superior	Rectus capitis posterior minor	
		Obliquus capitis superior	
		Scalenus anterior	
		Scalenus medius	
		Scalenus posterior	
		Longus colli	
		Rectus capitis anterior	
		Rectus capitis lateralis	

Temporomandibular joint				
Protraction	Retraction	Elevation	Depression	Lateropulsion
Lateral pterygoid	Temporalis (posterior fibres)	Temporalis (anterior fibers)	Mylohyoid	Medial pterygoid
Masseter (superficial part)		Masseter (superficial and deep part)	Digastricus	Lateral pterygoid
		Medial pterygoid	Geniohyoid	

Body part / joint	Movement	Range of joint motion
Scapula Acromioclavicular joint + Sternoclavicular joint + „Scapulothoracic joint“	depression – elevation	0–55°
	retraction – protraction	0–25°
	internal rotation – external rotation	0–60°
Arm / shoulder joint	ventral flexion	0–90° (180°)
	dorsiflexion / extension	0–50°
	abduction	0–90 (180°)
	adduction (hyperadduction)	0–75°
	external rotation	0–90°
	internal rotation	0–90°
Forearm / elbow joint	flexion	0–145°
	extension	0–5°
Forearm / proximal and distal radioulnar joints	supination	90°
	pronation	90°
Hand / wrist joint	palmar flexion	0–80°
	dorsal flexion	0–85°
	radial duction	0–15°
	ulnar duction	0–45°
Fingers of the hand / metacarpophalangeal joints	flexion	0–90°
	extension	0–20°
	abduction – adduction (from/to 3 rd finger)	0–45°
Thumb / carpometacarpal joint of first digit	flexion	0–45°
	extension	0–15°
	adduction – abduction	0–70°
Thigh / hip joint	flexion	0–130°
	extension	0–30°
	abduction	0–45°
	adduction (hyperadduction)	0–30°
	external rotation	0–45°
	internal rotation	0–35°
Leg / knee joint	extension– flexion	0–150°
	external rotation (in flexion only)	0–40°
	internal rotation (in flexion only)	0–10°
Foot / ankle joint	plantar flexion	0–50°
	dorsiflexion / extension	0–20°
Whole foot	inversion	0–30°
	eversion	0–15°
Fingers of the foot / metatarsophalangeal joints	flexion	0–45°
	extension	0–70°
	abduction – adduction (from/to 2 nd toe)	0–25°

Note 1: The range of joint motion is measured from 0° at the basic anatomical position to the maximal angle obtained for each particular movement. The exceptions are supination and pronation in which we describe the maximum range of motion to be 90° for both movements. The anatomical position of the forearm is in supination and therefore the total range of joint motion for supination and pronation should be 0–180°.

Note 2: In this table, values from goniometrical and kinesiological publications were used. These values vary widely. We tried to choose the most suitable compromise.

Glenohumeral joint					
Flexion	Extension	Abduction	Adduction	Internal rotation	External rotation
Deltoid (clavicular part)	Deltoid (spinal part)	Deltoid (acromial part)	Infraspinatus	Deltoid (clavicular part)	Deltoid (spinal part)
Biceps brachii (short head)	Teres major	Supraspinatus	Teres major	Teres major	Supraspinatus
Coracobrachialis	Triceps brachii (long head)	Biceps brachii (long head)	Teres minor	Subscapularis	Infraspinatus
Pectoralis major	Latissimus dorsi		Subscapularis	Latissimus dorsi	Teres minor
			Biceps brachii (short head)	Pectoralis major	
			Coracobrachialis		
			Triceps brachii (long head)		
			Latissimus dorsi		
			Pectoralis major		

Elbow joint			
Flexion	Extension	Pronation	Supination
Brachialis	Triceps brachii	Pronator teres	Supinator
Biceps brachii (only with supinated forearm)	Anconeus	Pronator quadratus	Biceps brachii (only with pronated forearm)
Palmaris longus		Brachioradialis (only in flexed forearm from supinated position)	Brachioradialis (only in extended forearm from pronated position)
Pronator teres		Extensor carpi radialis longus (only in flexed forearm)	Extensor carpi radialis longus (only in extended forearm)
Flexor carpi radialis		Extensor carpi radialis brevis (only in flexed forearm)	Extensor carpi radialis brevis (only in extended forearm)
Flexor carpi ulnaris			
Flexor digitorum superficialis			
Brachioradialis			

Wrist joint (and associated joints of the hand)			
Flexion	Extension	Radial duction	Ulnar duction
Flexor carpi radialis	Extensor carpi radialis longus	Flexor carpi radialis	Flexor carpi ulnaris
Flexor carpi ulnaris	Extensor carpi radialis brevis	Extensor carpi radialis longus	Extensor carpi ulnaris
Flexor digitorum superficialis	Extensor digitorum	Extensor carpi radialis brevis	
Flexor digitorum profundus	Extensor carpi ulnaris	Abductor pollicis longus	
Palmaris longus	Extensor digiti minimi		
Flexor pollicis longus	Extensor indicis		

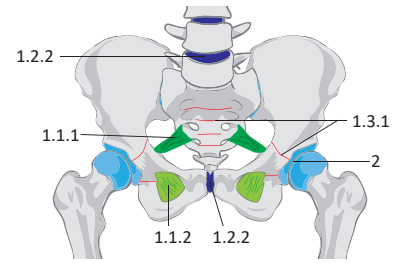
Hip joint					
Flexion	Extension	Abduction	Adduction	Internal rotation	External rotation
Iliopsoas	Gluteus maximus	Iliopsoas	Gluteus maximus	Gluteus medius (anterior fibers)	Iliopsoas
Gluteus medius (anterior fibers)	Gluteus medius (posterior fibers)	Gluteus maximus	Gracilis	Gluteus minimus (anterior fibers)	Gluteus maximus
Gluteus minimus (anterior fibers)	Gluteus minimus (posterior fibers)	Gluteus medius (middle part)	Pectineus	Tensor fasciae latae	Gluteus medius (posterior fibers)
Tensor fasciae latae	Adductor magnus (extensor part)	Gluteus minimus (middle part)	Adductor brevis	Semitendinosus	Gluteus minimus (posterior fibers)
Sartorius	Biceps femoris	Tensor fasciae latae	Adductor longus	Semimembranosus	Piriformis (in extended thigh)
Quadriceps femoris (rectus femoris)	Semitendinosus	Piriformis (in flexed thigh)	Adductor magnus (adductor part)	Adductor magnus (extensor part)	Gemellus superior (in extended thigh)
Pectineus	Semimembranosus	Gemellus superior (in flexed thigh)	Semitendinosus		Obturatorius internus (in extended thigh)
Adductor brevis		Obturator internus (in flexed thigh)	Semimembranosus		Gemellus inferior (in extended thigh)
Adductor longus		Gemellus inferior (in flexed thigh)			Quadratus femoris
Adductor magnus (adductor part)		Sartorius			Sartorius
					Pectineus
					Adductor brevis
					Adductor longus
					Biceps femoris

Knee joint			
Flexion	Extension	Internal rotation (only in flexed leg)	External rotation (only in flexed leg)
Sartorius	Quadriceps femoris	Sartorius	Biceps femoris
Gracilis	Gluteus maximus	Gracilis	
Biceps femoris	Tensor fasciae latae	Semitendinosus	
Semitendinosus		Semimembranosus	
Semimembranosus		Popliteus	
Triceps surae (gastrocnemius)			
Popliteus			

Ankle joint (and the associated joints of the foot)			
Flexion (plantar flexion)	Extension (dorsal flexion)	Inversion	Eversion
Fibularis longus	Tibialis anterior	Tibialis anterior	Fibularis longus
Fibularis brevis	Extensor digitorum longus	Tibialis posterior	Fibularis brevis
Triceps surae	Extensor hallucis longus	Flexor digitorum longus	Extensor digitorum longus
Tibialis posterior		Flexor hallucis longus	
Flexor digitorum longus		Triceps surae	
Flexor hallucis longus			

I. General arthrology

1. Explain the term "neutral position of a joint". (p. 68)
2. Explain the differences between synchondrosis and symphysis and give examples of both of these types of joints. (p. 68)
3. State what fills the articular cavity and what creates this substance. (p. 70)
4. Give the definitions of complex and compound joints. (p. 69)
5. Describe the differences between articular discs and menisci and give examples of where these structures can be found. (p. 71)
6. Explain the term "circumduction" and name some joints where this movement can be performed. (p. 71)



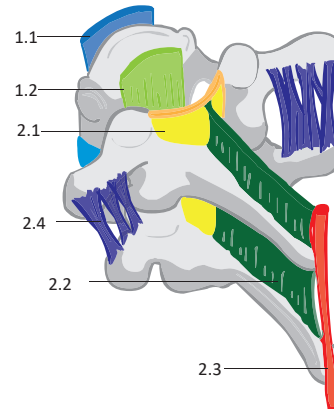
Describe the joints of the pelvis and hip

II. Joints of the skull

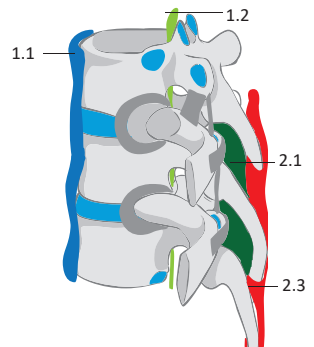
7. List all the types of joint found in the skull. (p. 72)
8. Describe the movements that can be performed in the temporomandibular joint. (p. 72)

III. Joints of the vertebral column

9. State the type of joint found between adjacent vertebral bodies and describe the structure of the intervertebral disc. (p. 73)
10. Name the two main long ligaments on the vertebral column and specify their location and their main function. (p. 73)
11. Name 5 ligaments of the median atlanto-axial joint. (p. 73)
12. Name the joint located between the adjacent superior and inferior articular processes of the vertebrae and state the accessory feature of this joint. (p. 73, 74)
13. Specify the part of the vertebral column that has the range of motion in rotation and specify the approximate range of this rotation. (p. 75)



List the ligaments of the spine



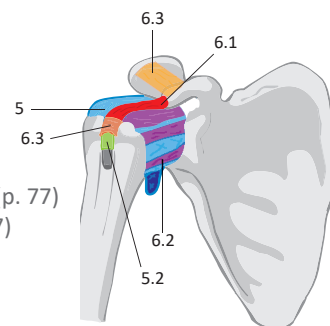
List the ligaments of the spine

IV. Thoracic joints

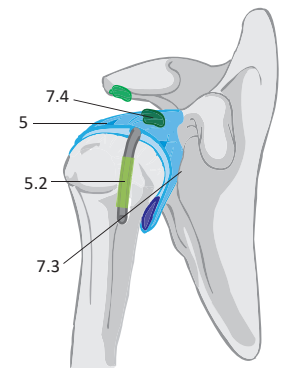
14. Name three main locations of connections of the true ribs to the rest of the skeleton. (p. 75)

V. Joints of the upper limb

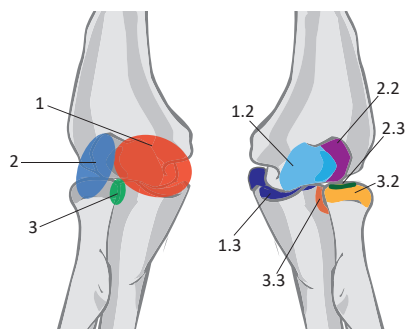
15. Specify the type of the acromioclavicular joint. (p. 76)
16. Describe the articular cavity of the glenohumeral joint. (p. 77)
17. Describe the movements that can be performed at the shoulder joint. (p. 77)
18. Name the muscle whose tendon runs through the shoulder joint. (p. 77)
19. Name one synovial bursa in the vicinity of the shoulder joint. (p. 77)
20. Name three joints that comprise the elbow joint. (p. 78)
21. Specify the shape of the proximal radioulnar joint. (p. 78)
22. Describe the neutral position of the elbow joint. (p. 78)
23. Name the three main ligaments of the elbow joint. (p. 78)
24. Describe the course of interosseous membrane and explain the term "oblique cord". (p. 79)
25. Describe the movements that can be performed in the proximal and distal radioulnar joint. (p. 79)
26. Specify the type and shape of the wrist joint and list the structures that articulate in it. (p. 79)
27. Name the articular head and fossa of the midcarpal joint. (p. 80)
28. Specify the shape of the metacarpophalangeal joints and describe the movements that can be performed in them. (p. 80)
29. Give an example of a saddle joint in the hand and describe the movements that can be performed in it. (p. 80)



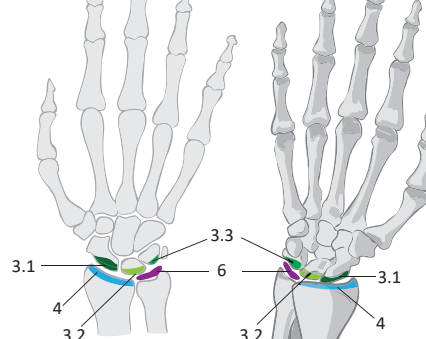
Describe the structures of the shoulder joint



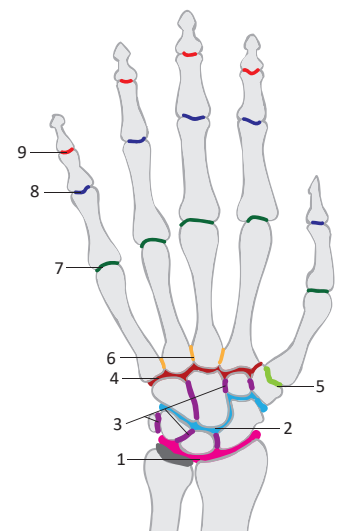
Describe the structures of the shoulder joint



Describe the components of the elbow joint



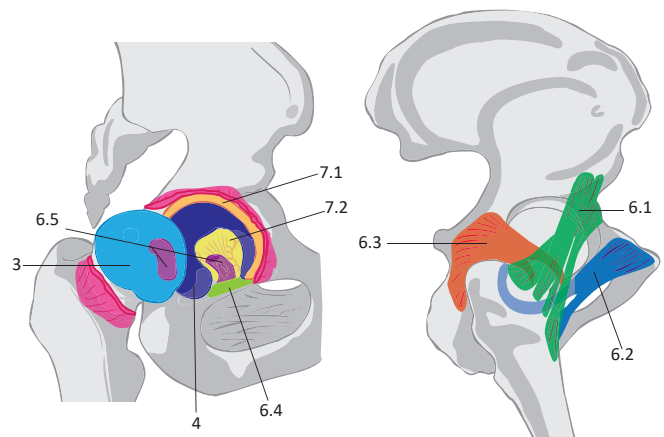
Describe the articular surfaces of the wrist joint



Describe the joints of the hand

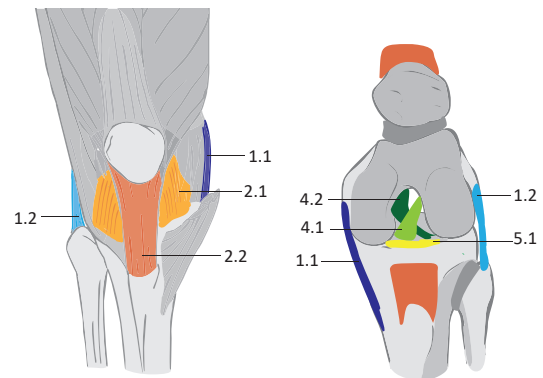
VI. Joints of the lower limb

30. State the name of the articular fossa of the hip joint. (p. 83)
31. Name the structure that enlarges the acetabulum and state which tissue it is made of. (p. 83)
32. Name the ligament that limits extension of the hip joint and specify where this ligament attaches on the femur. (p. 83)
33. Explain what is the "zona orbicularis" of the hip joint. (p. 83)
34. Describe the movements (including their range) that can be performed in the hip joint. (p. 83)
35. Explain the term "retinacula of Weitbrecht". (p. 83)
36. Name three synovial bursae in the vicinity of knee joint. (p. 84)
37. Name 4 intra-articular ligaments of the knee joint. (p. 85)
38. State the structure that attaches to the medial meniscus and limits its movability. (p. 85)
39. Describe flexion of the knee joint and name its three phases. (p. 85)
40. Describe the neutral position of the knee joint. (p. 85)
41. Describe the difference between the proximal and distal articulations of the tibia and fibula. (p. 86)
42. Specify the type and shape of the ankle joint. (p. 87)
43. Name 4 ligaments that form the medial ligament of the ankle joint. (p. 87)
44. Describe the "Chopart's joint". (p. 89)
45. Describe the bifurcate ligament (p. 89)
46. List the two rows of bones that form the medial and lateral longitudinal arches of the foot. (p. 89)
47. Name 4 muscles that support the transverse arch of the foot. (p. 89)
48. Name three muscles that support the longitudinal arch of the foot. (p. 89)

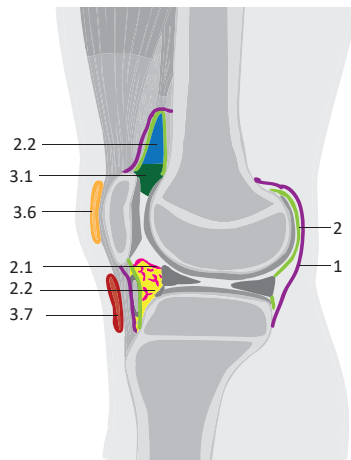


Name the components of the hip joint

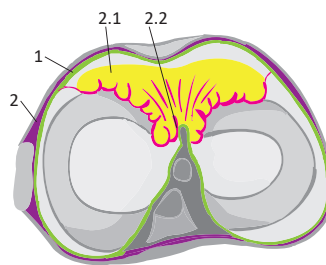
List the ligaments of the hip joint



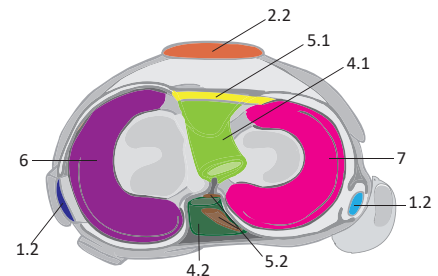
Name the ligaments of the knee joint



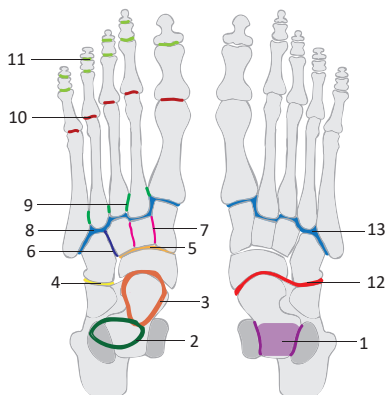
Describe the knee joint



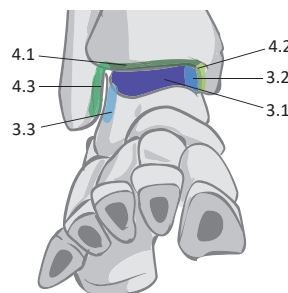
Describe the knee joint



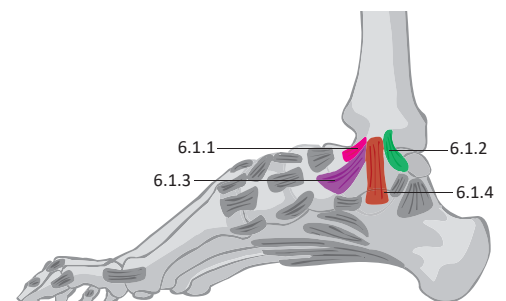
Describe the knee joint



Describe the joints of the foot



Describe the articular surfaces of the ankle joint



List the ligaments of the ankle joint

We would like to **thank the following anatomists, physicians and medical students** for their invaluable help, devotion and feedback in the preparation of this chapter.

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